

Combinatorial Coefficient Calculus

**Basis Changes, Labelled Structures, Special Polynomials,
Higher Composition, and Multivariate Inversion**

A unified proof-oriented monograph

September 2026

Abstract

This monograph is a unified, proof-oriented treatment of the coefficient arrays, polynomial families, and inversion principles that recur throughout enumerative combinatorics, formal power-series calculus, finite-difference theory, probability, and asymptotic analysis. It consolidates five overlapping draft manuscripts into one dependency-driven article, removes repeated exposition, reconciles conflicting normalizations, and incorporates additional material from the supplied Wikipedia and MathWorld dossiers together with primary mathematical sources.

The central objects are signed and unsigned Stirling numbers of the first kind, Stirling numbers of the second kind, Lah and generalized factorial coefficients, Bell numbers and Touchard polynomials, exponential and ordinary Bell polynomials, Eulerian numbers of ordinary, type- B , and second order, Bernoulli, Euler, Genocchi, Nörlund, and Cauchy–Bernoulli polynomials, Faà di Bruno composition coefficients, Sheffer sequences and exponential Riordan arrays, and the one-variable and multivariate Lagrange–Good inversion formulas. Their connections with finite differences, Möbius inversion, moments and cumulants, normal ordering, symmetric functions, gamma and polygamma functions, Barnes’ G -function, associahedra, rooted trees, Lambert W , Kepler’s equation, and Laplace- and Euler–Maclaurin-type asymptotics are developed systematically.

The emphasis is on derivation rather than cataloguing. Triangular recurrences are proved from insertions or basis changes; generating functions are obtained from labelled decompositions; inverse transforms are derived by Möbius inversion or matrix algebra; higher chain rules are proved from set partitions; implicit-series coefficients are proved by formal residues; and analytic versions are accompanied by explicit hypotheses, branch information, singularity diagnostics, or remainder bounds. Where a draft contained an incomplete proof, an index ambiguity, a suppressed Jacobian, or a schematic asymptotic placeholder, the statement has been repaired and replaced by a complete, checkable form.

How to read this work

The chapter order follows mathematical dependence rather than the order of the source files. Four recurring operations organize nearly all formulas:

1. change to a polynomial basis in which the relevant difference or derivative operator is simple;
2. decompose a labelled structure into connected components and translate that decomposition into an exponential generating function;
3. compose or invert formal series by Bell polynomials and residues; and
4. pass from exact coefficient formulas to analytic estimates by Cauchy, saddle-point, Euler–Maclaurin, Darboux, or Watson-type arguments.

Readers interested primarily in enumerative combinatorics may follow the Stirling, Bell, Eulerian, and associahedral chapters. Readers interested in symbolic calculus may proceed from Bell polynomials to Faà di Bruno, Sheffer–Riordan calculus, and Lagrange inversion. The Bernoulli–Euler and special-function part forms a parallel route from finite differences to summation and asymptotics. The appendices record source provenance, normalization conversions, reproducible algorithms, and independent consistency checks.

A formal power-series theorem is an algebraic statement: no convergence is required, and every coefficient depends on finitely many input coefficients. An analytic theorem is stated separately and includes the hypotheses needed for local inversion, termwise differentiation, contour deformation, Fourier convergence, or an asymptotic remainder estimate.

Editorial normalization

The consolidation applies the following conventions globally.

1. Bell numbers are B_n , complete exponential Bell polynomials are $\mathcal{B}_n$, partial exponential Bell polynomials are $\mathcal{B}_{n,k}$, and ordinary partial Bell polynomials are $\hat{\mathcal{B}}_{n,k}$.
2. Bernoulli numbers and polynomials are β_n and $\beta_n(x)$ with $\beta_1 = -1/2$. Euler polynomials $E_n(x)$ are distinguished from Eulerian polynomials and arrays.
3. Signed first-kind Stirling numbers are $s(n, k)$; unsigned first-kind numbers are $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$. The sign conversion is never left implicit.
4. Type- B descents are counted in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift present in one draft.
5. Ordinary and exponential coefficient conventions are linked explicitly by factorial conversions. Analytic inversion includes its Jacobian, branch, and regularity conditions; multivariate inversion includes Good’s determinant.

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Notation

The ground coefficient ring is usually a commutative $\mathbb{Q}$ -algebra; analytic statements specialize to $\mathbb{R}$ or $\mathbb{C}$. For $n \in \mathbb{N}$,

$$x^{\underline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\overline{n}} = x(x+1) \cdots (x+n-1),$$

with both values equal to 1 at $n = 0$. Coefficient extraction and formal residue notation are

$$[z^n]F(z) = [z^n]F(z), \quad \operatorname{Res}_z F(z) \, dz = [z^{-1}]F(z).$$

The principal triangular arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle, \quad \left\langle \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \right\rangle.$$

Bell numbers are B_n ; the complementary Bell number is $U_n = T_n(-1)$. Exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$, and $\widehat{\mathcal{B}}_{n,k}$ denotes the ordinary partial Bell polynomial. The Bernoulli convention is

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Unless stated otherwise, an impossible index range contributes zero, empty products equal 1, and $0^0 = 1$ in finite coefficient formulas. A bar such as $\bar{f}$ or the notation $f^{(-1)}$ denotes compositional inversion, not reciprocal inversion.

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,

$$[z^n]F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n]F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}]A(z)$. Then

$$[z^n]F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.2 (Formal change of variables for residues). Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,

$$\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + O(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$\hat{A}(z)\hat{B}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta x^{\underline{k}} = kx^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $x + 1^{\underline{k}} - x^{\underline{k}} = kx^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.3 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} x^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k x^{\underline{k}}$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n(x_1, \dots, x_n)$.

Theorem 1.4 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Corollary 1.5 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

1.6 Binomial convolution and Boolean-lattice inversion

The forward-difference formula is the one-dimensional shadow of Möbius inversion on subsets. It is convenient to record the full transform before specializing it repeatedly in later chapters.

Theorem 1.6 (Parameterized binomial inversion). *Let c be an element of a commutative ring and let $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$ be sequences. The relations*

$$b_n = \sum_{k=0}^n \binom{n}{k} c^{n-k} a_k \quad (1.14)$$

and

$$a_n = \sum_{k=0}^n \binom{n}{k} (-c)^{n-k} b_k \quad (1.15)$$

are equivalent. If $A(z) = \sum_{n \geq 0} a_n z^n / n!$ and $B(z) = \sum_{n \geq 0} b_n z^n / n!$, then both are equivalent to

$$B(z) = e^{cz} A(z). \quad (1.16)$$

Proof. Substitute (1.14) into the right side of (1.15). The coefficient of a_j becomes

$$\begin{aligned} \sum_{k=j}^n \binom{n}{k} (-c)^{n-k} \binom{k}{j} c^{k-j} &= \binom{n}{j} c^{n-j} \sum_{r=0}^{n-j} \binom{n-j}{r} (-1)^{n-j-r} \\ &= \binom{n}{j} c^{n-j} (1-1)^{n-j}, \end{aligned}$$

which is 1 for $j = n$ and 0 otherwise. This proves one implication; replacing c by $-c$ proves the converse. For the generating-function form, write

$$\begin{aligned} B(z) &= \sum_{n \geq 0} \sum_{k=0}^n \binom{n}{k} c^{n-k} a_k \frac{z^n}{n!} \\ &= \left(\sum_{r \geq 0} \frac{(cz)^r}{r!} \right) \left(\sum_{k \geq 0} a_k \frac{z^k}{k!} \right) = e^{cz} A(z). \end{aligned}$$

Multiplication by e^{-cz} gives the inverse transform. $\square$

Theorem 1.7 (Möbius inversion on the Boolean lattice). *Let 2^X be the subsets of a finite set X , ordered by inclusion. Its Möbius function is*

$$\mu(S, T) = (-1)^{|T \setminus S|} \quad (S \subseteq T). \quad (1.17)$$

Consequently, the relations

$$G(T) = \sum_{S \subseteq T} F(S) \quad (1.18)$$

and

$$F(T) = \sum_{S \subseteq T} (-1)^{|T \setminus S|} G(S) \quad (1.19)$$

are equivalent.

Proof. For $S \subseteq T$ the defining convolution identity for a Möbius function is

$$\sum_{S \subseteq U \subseteq T} \mu(S, U) = \mathbf{1}_{S=T}.$$

The proposed value gives

$$\sum_{S \subseteq U \subseteq T} (-1)^{|U \setminus S|} = \sum_{r=0}^{|T \setminus S|} \binom{|T \setminus S|}{r} (-1)^r = (1-1)^{|T \setminus S|},$$

so it is the Möbius function. Substitution of (1.18) into (1.19) and reversal of the finite sums now gives exactly the same convolution identity. $\square$

If $F(S)$ depends only on $|S|$, grouping subsets by cardinality turns Boolean Möbius inversion into ordinary binomial inversion. Thus inclusion–exclusion, finite differences, and the triangular transform in Theorem 1.6 are the same algebraic operation in three coordinate systems.

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$x^n = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$x^{\bar{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $x^{\bar{n}} = (-1)^n - x^n$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $x^{\bar{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n=3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x=1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). *For $n, k \geq 0$,*

$$\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). *For all $n, k \geq 0$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

The simple diagonals follow at once:

$$\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1, \quad \left\{ \begin{matrix} n \\ 1 \end{matrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{matrix} n \\ n-1 \end{matrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.6 (Alternative second-kind recurrences). *For $1 \leq k < n$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{matrix} n \\ k+j-1 \end{matrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{matrix} n-j+1 \\ k \end{matrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{matrix} n \\ r \end{matrix} \right\} k^r$. Since $k^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\binom{-k}{j} = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.7 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = y^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{y^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x) \log(1-x)) C'_k(x).$$

The left side has generating function $x C'_k - k C_k$. Their equality reduces to

$$-(1-x) \log(1-x) C'_k(x) = k C_k(x),$$

which follows immediately from the displayed formula for C_k . □

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1							0	1						
1	0	1						1	0	1					
2	0	1	1					2	0	1	1				
3	0	2	3	1				3	0	1	3	1			
4	0	6	11	6	1			4	0	1	7	6	1		
5	0	24	50	35	10	1		5	0	1	15	25	10	1	
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n \}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $x^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Corollary 3.2 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} x - 1^{\underline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} n^{\underline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $x^{\underline{k}} = x x - 1^{\underline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Corollary 3.3 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\{n_k\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand x^n into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{N+1^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2}N + 1^2 + \frac{7}{3}N + 1^3 + \frac{6}{4}N + 1^4 + \frac{1}{5}N + 1^5,$$

which expands to the familiar quartic power-sum polynomial.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.4 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}X^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{r^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \{n_k\} X^{\underline{k}}$. $\square$

Theorem 3.5 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n .

Proof. The random variable Y_m^k counts ordered k -tuples of distinct fixed points. There are m^k candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}Y_m^k = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}Y_m^k = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. $\square$

3.4 Lah numbers

Definition 3.6. For $1 \leq k \leq n$, the unsigned Lah number is

$$L(n, k) = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $L(0, 0) = 1$ and zero otherwise outside the triangle.

Combinatorially, $L(n, k)$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.7 (Conversion between rising and falling factorials).

$$x^{\bar{n}} = \sum_{k=0}^n L(n, k) x^k, \quad (3.16)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\bar{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} L(n, j) L(j, k) = \delta_{nk}. \quad (3.18)$$

Moreover,

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} L(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k L(n, k) x^k \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} x^{\bar{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq L(n, k) \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.2 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} 1/x^{k+1}}. \quad (4.6)$$

Consequently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\begin{Bmatrix} n \\ r \end{Bmatrix}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ by $k!$.

4.3 The Stirling transform

Theorem 4.3 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$\widehat{G}(z) = \widehat{F}(e^z - 1), \quad \widehat{F}(z) = \widehat{G}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.3. For the generating function,

$$\widehat{G}(z) = \sum_k f_k \sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = \widehat{F}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $m^{\underline{k}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.4 (Stirling normal-ordering identities). *For $n \geq 0$,*

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = x D^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^{\underline{n}} = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} m^{\underline{k}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $m^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, $2 + 2$ for the second diagonal and 4, $3 + 2$, $2 + 2 + 2$ for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m[x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ n \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ j \end{matrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx} x^{\bar{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X+c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.2 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j-k$ and lower index $j-n$:

$$\begin{Bmatrix} j-k \\ j-n \end{Bmatrix} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-m-n+k)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.3 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{p \prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p=0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p=0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1+A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$x^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $x^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $x^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

6.2 Parity of second-kind numbers

Theorem 6.2 (Parity criterion). *Let*

$$z = n - \left\lceil \frac{k+1}{2} \right\rceil, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv \binom{z}{w} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\left\{ \begin{matrix} 2n \\ n \end{matrix} \right\}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\binom{n-k+\lceil k/2 \rceil-1}{\lceil k/2 \rceil-1} = \binom{z}{w}$. Lucas's theorem modulo 2 says $\binom{z}{w}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \left\{ \begin{matrix} -k \\ -n \end{matrix} \right\}, \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left[\begin{matrix} -k \\ -n \end{matrix} \right]. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\left[\begin{matrix} -n \\ k \end{matrix} \right] = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\left[\begin{matrix} -5 \\ k \end{matrix} \right] = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x)\cdots(n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$B_{-1} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$. $\square$

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = [t^k] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $O(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$T_{n+1}(x) = x(T_n(x) + T'_n(x)), \quad T_0(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of T_n are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $T_1(x) = x$. Suppose $f = T_n$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + O(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + O(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

Let $v = n/k > 1$ and let $G \in (0, 1)$ be the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Equivalently, if $r = v - G$, then $r/(1 - e^{-r}) = v$. Saddle-point extraction from $(e^z - 1)^k/k!$ gives

$$\left\{ \frac{n}{k} \right\} \sim \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k}, \quad (7.13)$$

uniformly away from the two extreme edges; refined versions have relative error $O(1/n)$, and the numerical constant $0.066/n$ quoted in the archive belongs to one particular explicit error theorem.

Derivation. By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi i k!} \int \exp(k \log(e^z - 1) - n \log z) \frac{dz}{z}.$$

The positive saddle r solves $ke^r/(e^r - 1) = n/r$, or $r/(1 - e^{-r}) = v$. Put $G = v - r$ to obtain Equation (7.12). Expanding the phase to second order at r , evaluating the Gaussian integral, and simplifying with Stirling's formula for $n!/k!$ gives exactly Equation (7.13). On a fixed angular complement of the saddle the real part of the phase drops quadratically; Taylor's theorem with an explicit third- and fourth-derivative bound gives a relative $O(1/n)$ remainder. Thus the formula is a theorem of the saddle method, not a heuristic replacement of the generating function. $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.4 (Stirling-zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n v^{\bar{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i / (e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/v^{\bar{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{v^{\bar{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m} / (e^u - 1) du = \Gamma(z+1-m) \zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\zeta(s, v) = \frac{k!}{s - k} \sum_{n \geq 0} \frac{\begin{bmatrix} n+k \\ n \end{bmatrix}}{(n+k)!} \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.5 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $O(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^w$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z+w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.6 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.22)$$

Proof. Put $L(u) = -\log(1-u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell+1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right], \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \left[\begin{matrix} n \\ r \end{matrix} \right] u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1-u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.5 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell+1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.7 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m = & \frac{1}{2}\delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ & \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^{\infty} \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^{\infty} \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1+w)}{1+w} = m! \sum_{r \geq m} s(r+1, m+1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k+2, m+1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k+1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k+1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k+1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.5 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno’s formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n are defined by

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} G_n z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1+z)$ gives

$$n!G_n = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell+1}. \quad (7.30)$$

Indeed, $z/\log(1+z) = \int_0^1 (1+z)^t dt$ and comparison with $(1+z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n \mu^k s(n, n-k+1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}D_t^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand D_t^n by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N} \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := [x^{k-1}](x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.2 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n + 1$. If deleting $n + 1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n + 1$ has exactly r elements: choose its $r - 1$ companions and partition the remaining $n - r + 1$ labels into $k - 1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j / j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.3 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n - d + 1 \\ k - d + 1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i - j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) q^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d - 1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d - 1$ vertices, which form a clique, and hence has $q - d + 1$ choices. Thus

$$\chi_{G_{n,d}}(q) = q^{d-1} (q - d + 1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned} \chi_{G_{n,d}}(q) &= q^{d-1} \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q - d + 1^r \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r+d-1}. \end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). $\square$

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0 = B_1 = 1, \quad (B_n)_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.4. *If $N = p_1 \cdots p_n$ is squarefree, then B_n is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.5. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.6. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.7 (Return-to-order shuffles). *Exactly B_n of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.8. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.9. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{ccccccc} & & & & & & 1 \\ & & & & & & 1 & 2 \\ & & & & & 2 & 3 & 5 \\ & & & 5 & 7 & 10 & 15 \\ 15 & 20 & 27 & 37 & 52. \end{array}$$

Theorem 9.10. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n$ and the right edge is $a_{n,n} = B_{n+1}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). For $n \geq 0$,

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

10.2 Spivey's identity and binomial inversions

Theorem 10.3 (Spivey's identity). For $m, n \geq 0$,

$$B_{m+n} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.4 (Bell inversion identities). For $n, k \geq 0$,

$$B_n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.7 below gives $\mathbb{E}X^r = B_r$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.5 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}. \quad (10.9)$$

Proof. By Dobiński's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} m^i = m^{\underline{k}}$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{m^{\underline{k}} [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

10.3 Exponential generating function and differential equation

Theorem 10.6. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

10.4 Dobiński's formula and Poisson moments

Theorem 10.7 (Dobiński). *For every $n \geq 0$,*

$$B_n = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z-1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a $\text{Poisson}(1)$ variable. $\square$

Corollary 10.8 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. $\square$

10.5 Complementary Bell numbers and parity-weighted partitions

The MathWorld dossier includes the complementary Bell, or Uppuluri–Carpenter, numbers. They fit naturally into the Touchard-polynomial calculus and require no separate machinery.

Definition 10.9. Define

$$U_n := T_n(-1) = \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (10.13)$$

Thus U_n is the number of partitions of $[n]$ with an even number of blocks minus the number with an odd number of blocks. The first values are

$$1, -1, 0, 1, 1, -2, -9, -9, 50, 267, 413, \dots$$

Theorem 10.10. *The complementary Bell numbers satisfy*

$$\sum_{n \geq 0} U_n \frac{z^n}{n!} = \exp(1 - e^z), \quad (10.14)$$

$$U_n = e \sum_{m=0}^{\infty} \frac{(-1)^m m^n}{m!}, \quad (10.15)$$

$$U_{n+1} = - \sum_{k=0}^n \binom{n}{k} U_k. \quad (10.16)$$

Proof. The bivariate Touchard generating function is

$$\sum_{n \geq 0} T_n(x) \frac{z^n}{n!} = \exp(x(e^z - 1)).$$

Setting $x = -1$ proves (10.14). Expanding the outer exponential gives

$$e^{1-e^z} = e \sum_{m \geq 0} \frac{(-1)^m e^{mz}}{m!},$$

and coefficient comparison proves (10.15); the series is absolutely convergent for each fixed n . Finally, if $C(z) = e^{1-e^z}$, then $C'(z) = -e^z C(z)$. Comparing coefficients of $z^n/n!$ gives (10.16). $\square$

More generally, $T_n(x)$ is the partition enumerator in which every block has weight x . Ordinary Bell numbers and complementary Bell numbers are therefore the evaluations $x = 1$ and $x = -1$ of the same polynomial family; the latter measure the exact cancellation produced by block parity.

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n to the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n(1) = B_n.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}(x) \equiv T_{n+1}(x) + x^p T_n(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p} \equiv B_n + B_{n+1} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Theorem 11.2 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m} \equiv m B_n + B_{n+1} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}(x) \equiv T_{n+1}(x) + T_n(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1} + T_n \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

11.2 A universal period bound modulo a prime

Theorem 11.3. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.4. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.5. *For $n \geq 1$,*

$$1 \leq \frac{B_{n-1}B_{n+1}}{B_n^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n) is log-convex, whereas $(B_n/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1}\mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{t-1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1} z_{j+1}) = j/(j-1)$. Hence $(B_n/n!)^2 \geq (B_{n-1}/(n-1)!)(B_{n+1}/(n+1)!)$, which is the right inequality in (11.6). $\square$

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.6. *For $t > 0$,*

$$B_n \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$B_n^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.7 (Uniform numerical bounds). *For every $n \geq 1$,*

$$B_n < \left(\frac{0.792 n}{\log(n+1)} \right)^n. \quad (11.9)$$

Define, for $x > 1$,

$$d(x) = \log \log(x+1) - \log \log x + \frac{1 + e^{-1}}{\log x}. \quad (11.10)$$

For every $\varepsilon > 0$ there is $n_0(\varepsilon) = \max\{e^4, d^{-1}(\varepsilon)\}$ such that, for $n > n_0(\varepsilon)$,

$$B_n < \left(\frac{e^{-0.6+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.11)$$

Proof. Set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

Direct differentiation, after multiplying by the positive denominator $2x \log x \log(x+1) \log(x/\log(x+1))^2$, shows that $R'(x) < 0$ for $x \geq 39$; substitution gives $R(39) < 0.792$. Thus (11.8) proves (11.9) for $n \geq 39$. The remaining $1 \leq n \leq 38$ are obtained successively from (10.1); substituting them gives strict inequalities. This is a finite exact integer check.

A second differentiation gives, for $x \geq e^4$,

$$\log R(x) \leq -0.6 + d(x), \quad d'(x) < 0, \quad \lim_{x \rightarrow \infty} d(x) = 0.$$

(The numerator after clearing positive denominators is a sum of $-(\log x - 4)^2$, $-(x - 1 - \log x)$, and negative multiples of these two standard nonnegative quantities.) Hence $d(n) < \varepsilon$ when $n > d^{-1}(\varepsilon)$, and (11.11) follows from (11.8). $\square$

11.5 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1+r)$.

Theorem 11.8 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \quad (11.12)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1\right). \quad (11.13)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n}{n!} = \frac{e^{e^r} - 1}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du/\sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.12). Stirling's formula and $e^r = n/r$ simplify it to (11.13). $\square$

The archived refined expansion used undefined symbols. The following construction makes every coefficient unambiguous. Let $T_j(r) = \sum_{q=0}^j \{j\}_q r^q$ be the Touchard polynomial and let $Z \sim N(0, 1)$. For an integer $h = O(r)$ define

$$\mathcal{R}_{r,h}(Z) = \exp\left(-\frac{ihZ}{\sqrt{A}} + \sum_{j \geq 3} \frac{i^j e^r T_j(r)}{j! A^{j/2}} Z^j\right). \quad (11.14)$$

Expand this formal exponential and take Gaussian moments termwise; retaining all monomials of total order at most e^{-Mr} defines $C_0(r, h), \dots, C_M(r, h)$, with $C_0 = 1$.

Theorem 11.9 (Uniform Moser–Wyman scheme). *For every fixed M and $|h| \leq Cr$,*

$$B_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi A}} \left(\sum_{q=0}^{M-1} C_q(r, h) + O_C(e^{-Mr} P_M(r)) \right), \quad (11.15)$$

where P_M is an explicitly computable polynomial. The first correction is

$$C_1(r, h) = -\frac{h^2}{2A} - \frac{he^r T_3(r)}{2A^2} + \frac{e^r T_4(r)}{8A^2} - \frac{5e^{2r} T_3(r)^2}{24A^3}. \quad (11.16)$$

In particular the coefficient of e^{-qr} is a polynomial in h of degree at most $2q$, with rational functions of r as coefficients.

Proof. Use the same circle as in the preceding proof but replace n by $n+h$. After $\theta = Z/\sqrt{A}$, the ratio of the integrand to its Gaussian quadratic part is exactly the formal series (11.14). Watson's lemma with a Gaussian majorant permits termwise integration on the central arc; the complementary arcs have the same exponential bound uniformly for $|h| \leq Cr$. Odd Gaussian moments vanish, so only integral powers of e^{-r} remain. Since $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, the terms from the square of the h -term, its product with the cubic term, the quartic term, and the square of the cubic term give (11.16). The same finite bookkeeping defines every subsequent C_q and bounds the omitted Taylor remainder. $\square$

Corollary 11.10 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n}{n} &= \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ &\quad + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.17)$$

Proof. Take logarithms in (11.13). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + r \log \log n \log n + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $O(\log n/n)$. $\square$

11.6 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0 = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\hat{\mathcal{B}}_{n,k}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\hat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $\hat{X}(t) = \sum_{j \geq 1} x_j t^j$,

$$\hat{X}(t)^k = \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{u\hat{X}(t)} = \sum_{n \geq k \geq 0} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1} &= x_1, \\
\mathcal{B}_{2,1} &= x_2, & \mathcal{B}_{2,2} &= x_1^2, \\
\mathcal{B}_{3,1} &= x_3, & \mathcal{B}_{3,2} &= 3x_1x_2, & \mathcal{B}_{3,3} &= x_1^3, \\
\mathcal{B}_{4,1} &= x_4, & \mathcal{B}_{4,2} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3} &= 6x_1^2x_2, & \mathcal{B}_{4,4} &= x_1^4, \\
\mathcal{B}_{5,1} &= x_5, & \mathcal{B}_{5,2} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4} &= 10x_1^3x_2, & \mathcal{B}_{5,5} &= x_1^5, \\
\mathcal{B}_{6,1} &= x_6, & \mathcal{B}_{6,2} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5} &= 15x_1^4x_2, & \mathcal{B}_{6,6} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0 &= 1, \\
\mathcal{B}_1 &= x_1, \\
\mathcal{B}_2 &= x_1^2 + x_2, \\
\mathcal{B}_3 &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4 &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5 &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6 &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7 &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}, \quad (13.2)$$

with $\mathcal{B}_{0,0} = 1$, $\mathcal{B}_{n,0} = 0$ for $n > 0$, and $\mathcal{B}_{0,k} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}(x) \mathcal{B}_{n-i,k_2}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B (|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1} = x_n, \quad \mathcal{B}_{n,2} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n} = x_1^n, \quad \mathcal{B}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (1.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.5. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t)^2 F(t)).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}(1, 1, \dots, 1) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (14.3)$$

$$\mathcal{B}_n(1, 1, \dots, 1) = \mathcal{B}_n, \quad (14.4)$$

$$\mathcal{B}_{n,k}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = L(n, k), \quad (14.5)$$

$$T_n(x) = \mathcal{B}_n(x, x, \dots, x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $T_n(x)$. $\square$

14.2 Homogeneity and a useful evaluation

Theorem 14.2 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}(1, 2, 3, \dots, n-k+1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k}k^{n-k}$. $\square$

14.3 Addition and shifted-zero identities

Theorem 14.3 (Complete Bell addition formula).

$$\mathcal{B}_n(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}(x) \mathcal{B}_i(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.4 (Insertion of q leading zeros). For $q \geq 0$,

$$\begin{aligned} \mathcal{B}_{n,k} \left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots \right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

14.4 Exponential–logarithmic inversion

Theorem 14.5 (Bell transform and its inverse). Suppose

$$y_n = \mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n/n!$. $\square$

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno's formula in Equation (24.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n = (h_{ij})$ and $K_n = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1} x_2 & \binom{n-1}{2} x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1} x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.6 (Bell determinants).

$$\mathcal{B}_n(x_1, \dots, x_n) = \det H_n = \det K_n. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n = D(JH_n^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.7. *If $x^{k\Diamond}$ denotes the k -fold diamond power, then*

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k\Diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k\Diamond}$ is $X(t)^k$. Compare with (1.10). □

For example,

$$x\Diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3\Diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$.

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n \sum_{k=1}^n \frac{1}{k} \widehat{\mathcal{B}}_{n,k}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k = -(k-1)! \operatorname{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n(s_1, s_2, \dots, s_n). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \operatorname{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.5. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n , $\mathcal{B}_n$, $\beta_n(x)$, and C_n are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$A_0(t) = 1, \quad A_n(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$A_n(t) = (1 + (n-1)t)A_{n-1}(t) + t(1-t)A'_{n-1}(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1 - \pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad A_n(t) = t^{n-1} A_n(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} A_n(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.4. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Theorem 16.5 (A second polynomial recurrence). For $n > 1$,

$$A_n(t) = \sum_{k=0}^{n-1} \binom{n}{k} A_k(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

16.3 Worpitzky's identity and sums of powers

Theorem 16.6 (Worpitzky). For every indeterminate x and $n \geq 1$,

$$x^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Corollary 16.7 (Power sums). For $m, n \geq 1$,

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z A_n(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \right\rangle_k z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.8.

$$A_n(t) = \sum_{k=0}^n \left\{ n \right\}_k k! (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\{ n \right\}_k m^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} m^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $A_n(t) = \sum_k \left\{ n \right\}_k k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.9. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle n \right\rangle_k f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0, 1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle n \right\rangle_k / n!$.

Proof. The volume of $\{x \in [0, 1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle n \right\rangle_k$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.10. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n}{k}} = (n+1) \beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n A_n \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $A_n(t)$, so its h^* -vector is the n th Eulerian row.*

The regions $\Delta_{n,k}$ in Theorem 16.9 are half-open hypersimplices. Their normalized volumes are $n! \operatorname{vol} \Delta_{n,k} = \langle \begin{smallmatrix} n \\ k \end{smallmatrix} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ordered partitions into k blocks.

Theorem 17.2. *The h -polynomial of the simple polytope dual to the type- A permutohedron is $A_n(t)$.*

Proof. For a simple $(n-1)$ -polytope the h -polynomial is the transform $h(t) = \sum_F (t-1)^{\dim F}$ over faces of the dual simplicial complex. Grouping braid-fan faces by their k ordered blocks gives $\sum_k k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} (t-1)^{n-k}$, which is $A_n(t)$ by (16.12). $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_B(n, k)$ count signed permutations with k such descents, and set $M_n(t) = \sum_{k=0}^n E_B(n, k)t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_B(n, k) = (2k + 1)E_B(n - 1, k) + (2n - 2k + 1)E_B(n - 1, k - 1), \quad (18.1)$$

$$\sum_{m \geq 0} (2m + 1)^n t^m = \frac{M_n(t)}{(1 - t)^{n+1}}, \quad (18.2)$$

$$E_B(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j) + 1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n - 1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k + 1$ preserving slots for an object with k descents and $2n - 2k + 1$ creating slots for one with $k - 1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m + 1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m + 1)^n = \sum_{k=0}^n E_B(n, k) \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1 - t)^{n+1}$ to obtain (18.2). Multiplication by $(1 - t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

As in type A , these coefficients form the h -vector of the simple polytope dual to the type- B permutohedron; the proof repeats the braid-fan argument with signed ordered partitions.

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$P_n(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad P_0(t) = 1.$$

Theorem 18.2 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n-1 \rangle\rangle_{k-1} + (k+1) \langle\langle n-1 \rangle\rangle_k, \quad (18.4)$$

$$P_{n+1}(t) = (2nt + 1)P_n(t) - t(t-1)P'_n(t), \quad (18.5)$$

$$P_n(1) = (2n-1)!! \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n-1$ gaps of a Stirling permutation of order $n-1$. A descent gap or the terminal gap preserves the number of descents, giving $k+1$ choices from an object with k descents. Each of the remaining $2n-k-1$ gaps creates one descent from an object with $k-1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n-1$ insertion gaps, so the total count is $(2n-1)!!$. $\square$

The differential recurrence has two useful equivalent forms:

$$(t-1)^{-2n-2}P_{n+1}(t) = (t(1-t)^{-2n-1}P_n(t))', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1-t} u_n(t) \right)', \quad u_n(t) = (t-1)^{-2n} P_n(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.3 (Diagonal Stirling generating function). For $n \geq 1$,

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n+m \\ m \end{matrix} \right\} t^m = \frac{tP_n(t)}{(1-t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1-t)F_{n+1} = tF'_n$, with $F_0 = (1-t)^{-1}$. Hence $F_1 = t/(1-t)^3$. If the displayed formula holds for n , differentiating $tP_n/(1-t)^{2n+1}$ and using (18.5) gives the formula for $n+1$. $\square$

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.4 (Second-order Eulerian diagonal formula). For $p, m \geq 0$,

$$\left[\begin{matrix} m \\ m-p \end{matrix} \right] = \sum_{j=0}^p \langle\langle p \rangle\rangle_j \binom{m+j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\begin{bmatrix} x \\ x-p \end{bmatrix} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \begin{bmatrix} x \\ x-1 \end{bmatrix} &= \binom{x}{2}, \\ \begin{bmatrix} x \\ x-2 \end{bmatrix} &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \begin{bmatrix} x \\ x-3 \end{bmatrix} &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \begin{bmatrix} m \\ m-p \end{bmatrix} t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} P_p(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2 ((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} P_p(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $P_p(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Bernoulli, Euler, and Summation Calculus

Chapter 19

Bernoulli numbers, Bernoulli polynomials, and Nörlund extensions

19.1 Generating functions and first consequences

The Bernoulli numbers have already appeared in the Eulerian alternating sums. We now develop their own coefficient calculus. Our convention is $\beta_1 = -1/2$, consistent with the generating-function normalization used in the DLMF [13].

Definition 19.1. The Bernoulli polynomials $\beta_n(x)$ are defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} \beta_n(x) \frac{t^n}{n!}, \quad \beta_n := \beta_n(0). \quad (19.1)$$

The first values are

$$\begin{aligned} \beta_0(x) &= 1, \\ \beta_1(x) &= x - \frac{1}{2}, \\ \beta_2(x) &= x^2 - x + \frac{1}{6}, \\ \beta_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{1}{2}x, \\ \beta_4(x) &= x^4 - 2x^3 + x^2 - \frac{1}{30}, \\ \beta_5(x) &= x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x. \end{aligned}$$

Theorem 19.2 (Appell, addition, difference, and reflection laws). *For $n \geq 0$,*

$$\beta'_n(x) = n\beta_{n-1}(x), \quad (19.2)$$

$$\beta_n(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k(x) y^{n-k}, \quad (19.3)$$

$$\beta_n(x+1) - \beta_n(x) = nx^{n-1} \quad (n \geq 1), \quad (19.4)$$

$$\beta_n(1-x) = (-1)^n \beta_n(x). \quad (19.5)$$

In particular,

$$\beta_n(x) = \sum_{k=0}^n \binom{n}{k} \beta_k x^{n-k}. \quad (19.6)$$

Moreover, $\beta_{2m+1} = 0$ for every $m \geq 1$.

Proof. Differentiate (19.1) with respect to x ; coefficient comparison gives (19.2). Multiplication of the generating function by e^{yt} gives (19.3); setting $x = 0$ gives (19.6). Next,

$$\frac{te^{(x+1)t}}{e^t - 1} - \frac{te^{xt}}{e^t - 1} = te^{xt},$$

whose coefficient of $t^n/n!$ is nx^{n-1} for $n \geq 1$. Finally,

$$\frac{te^{(1-x)t}}{e^t - 1} = \frac{(-t)e^{x(-t)}}{e^{-t} - 1},$$

so replacing t by $-t$ proves (19.5). At $x = 0$ this says $\beta_n(1) = (-1)^n \beta_n$. For $n \neq 1$ the difference law gives $\beta_n(1) = \beta_n$, hence odd β_n vanish for $n > 1$. $\square$

19.2 Faulhaber summation and the multiplication theorem

Theorem 19.3 (Faulhaber's formula in translated form). *For $p \geq 0$, an integer $N \geq 0$, and arbitrary x ,*

$$\sum_{r=0}^{N-1} (x+r)^p = \frac{\beta_{p+1}(x+N) - \beta_{p+1}(x)}{p+1}. \quad (19.7)$$

Proof. By (19.4),

$$\frac{\beta_{p+1}(u+1) - \beta_{p+1}(u)}{p+1} = u^p.$$

Set $u = x + r$ and sum from $r = 0$ to $N - 1$; the left side telescopes. $\square$

Thus the Bernoulli basis is adapted to discrete antidifferentiation just as the falling-factorial basis is adapted to the forward-difference operator.

Theorem 19.4 (Raabe multiplication formula). *For every positive integer q ,*

$$\sum_{r=0}^{q-1} \beta_n \left(x + \frac{r}{q} \right) = q^{1-n} \beta_n(qx). \quad (19.8)$$

Proof. Sum the generating functions and use the finite geometric series:

$$\begin{aligned} \sum_{r=0}^{q-1} \frac{te^{(x+r/q)t}}{e^t - 1} &= \frac{te^{xt}}{e^t - 1} \frac{e^t - 1}{e^{t/q} - 1} \\ &= q \frac{(t/q)e^{(qx)(t/q)}}{e^{t/q} - 1}. \end{aligned}$$

The coefficient of $t^n/n!$ on the right is $q^{1-n} \beta_n(qx)$. $\square$

19.3 A Stirling-number formula

Theorem 19.5. *For $n \geq 0$,*

$$\beta_n = \sum_{k=0}^n (-1)^k \frac{k!}{k+1} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (19.9)$$

Proof. The formal identity

$$\sum_{k \geq 0} \frac{(-1)^k u^k}{k+1} = \frac{\log(1+u)}{u}$$

with $u = e^t - 1$ becomes

$$\frac{t}{e^t - 1} = \sum_{k \geq 0} \frac{(-1)^k}{k+1} (e^t - 1)^k.$$

Now use $(e^t - 1)^k = k! \sum_{n \geq k} \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} t^n / n!$ and compare coefficients. Only $k \leq n$ contributes, so every coefficient calculation is finite. $\square$

This formula is also the integral identity

$$\beta_n = \int_0^1 T_n(-u) \, du,$$

because integration of u^k supplies the factor $1/(k+1)$. It places Bernoulli numbers, Touchard polynomials, and Stirling numbers in one triangular transform.

19.4 Generalized Bernoulli polynomials

Definition 19.6. For a scalar parameter α , define the generalized Bernoulli, or Nörlund, polynomials by

$$\left(\frac{t}{e^t - 1} \right)^\alpha e^{xt} = \sum_{n \geq 0} \beta_n^{(\alpha)}(x) \frac{t^n}{n!}. \quad (19.10)$$

Thus $\beta_n^{(1)}(x) = \beta_n(x)$ and $\beta_n^{(0)}(x) = x^n$.

Theorem 19.7 (Convolution calculus). *For arbitrary parameters α, γ ,*

$$\frac{\partial}{\partial x} \beta_n^{(\alpha)}(x) = n \beta_{n-1}^{(\alpha)}(x), \quad (19.11)$$

$$\beta_n^{(\alpha+\gamma)}(x+y) = \sum_{k=0}^n \binom{n}{k} \beta_k^{(\alpha)}(x) \beta_{n-k}^{(\gamma)}(y). \quad (19.12)$$

Furthermore,

$$\beta_{n-1}^{(n)} = (-1)^{n-1} (n-1)! \quad (n \geq 1). \quad (19.13)$$

Proof. The derivative and convolution laws follow respectively by differentiating and multiplying the generating functions. For the diagonal value, Lagrange inversion applied to the inverse pair $e^t - 1$ and $\log(1+z)$ gives

$$[t^{n-1}] \left(\frac{t}{e^t - 1} \right)^n = (-1)^{n-1}.$$

By (19.10), the left side is $\beta_{n-1}^{(n)} / (n-1)!$. $\square$

The complete Bell polynomials give an explicit construction. From

$$\log \frac{t}{e^t - 1} = -\frac{t}{2} - \sum_{r \geq 1} \frac{\beta_{2r}}{2r(2r)!} t^{2r}, \quad (19.14)$$

we obtain

$$\beta_n^{(\alpha)}(x) = \mathcal{B}_n(c_1, c_2, \dots, c_n), \quad (19.15)$$

where

$$c_1 = x - \frac{\alpha}{2}, \quad c_{2r} = -\frac{\alpha\beta_{2r}}{2r}, \quad c_{2r+1} = 0 \quad (r \geq 1).$$

Indeed, the logarithm of the generating function in (19.10) is $\sum_{j \geq 1} c_j t^j / j!$, and the complete Bell polynomial is the exponential of exactly such a series. Formula (19.14) follows by differentiating both sides, using the generating function for β_n , and matching the value at $t = 0$.

19.5 Fourier expansion and the even zeta values

Let $\tilde{\beta}_n(x) = \beta_n(\{x\})$ be the periodic Bernoulli function, where $\{x\}$ is the fractional part.

Theorem 19.8 (Fourier series of periodic Bernoulli polynomials). *For $n \geq 2$ and real x ,*

$$\tilde{\beta}_n(x) = -\frac{n!}{(2\pi i)^n} \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{2\pi i k x}}{k^n}. \quad (19.16)$$

The series is absolutely and uniformly convergent. Consequently, for $m \geq 1$,

$$\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{2(2m)!} \beta_{2m}. \quad (19.17)$$

Proof. For $k \neq 0$, let $c_k = \int_0^1 \beta_n(x) e^{-2\pi i k x} dx$. Since $\beta_r(1) = \beta_r(0)$ for $r \neq 1$, integration by parts and (19.2) give

$$c_k = \frac{n}{2\pi i k} c_k^{(n-1)} = \dots = \frac{n!}{(2\pi i k)^{n-1}} \int_0^1 \beta_1(x) e^{-2\pi i k x} dx.$$

One final integration by parts uses $\beta_1(1) - \beta_1(0) = 1$ and yields $c_k = -n!/(2\pi i k)^n$. The mean coefficient is zero because $\int_0^1 \beta_n(x) dx = 0$ for $n \geq 1$, by (19.2). Since $\sum_{k \neq 0} |k|^{-n}$ converges for $n \geq 2$, the Fourier series converges absolutely and uniformly to the continuous periodic function $\tilde{\beta}_n$. Setting $x = 0$ and $n = 2m$ gives

$$\beta_{2m} = -\frac{(2m)!}{(2\pi i)^{2m}} 2\zeta(2m),$$

which rearranges to (19.17). □

19.6 A modified even Bernoulli sequence

One of the supplied MathWorld notes records the coefficients $\mathfrak{b}_n$ defined by

$$\sum_{n \geq 0} \mathfrak{b}_n x^n = \frac{1}{2} \log \left(\frac{e^{x/2} - e^{-x/2}}{x/2} \right). \quad (19.18)$$

The constant term is $\mathfrak{b}_0 = \frac{1}{2} \log 2$, all positive odd coefficients vanish, and for even $n \geq 2$,

$$\mathfrak{b}_n = \frac{\beta_n}{2n n!}. \quad (19.19)$$

To prove this, differentiate (19.18) and use

$$\coth u = \frac{1}{u} + \sum_{r \geq 1} \frac{2^{2r} \beta_{2r}}{(2r)!} u^{2r-1}.$$

The singular terms cancel and termwise integration, with the value at $x = 0$, gives (19.19). Thus this “modified” sequence is a simple ordinary-generating normalization of the even Bernoulli numbers.

Chapter 20

Euler polynomials and Genocchi numbers

20.1 Euler polynomials

Definition 20.1. The Euler polynomials $E_n(x)$ are defined by

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} E_n(x) \frac{t^n}{n!}. \quad (20.1)$$

The first few are

$$\begin{aligned} E_0(x) &= 1, \\ E_1(x) &= x - \frac{1}{2}, \\ E_2(x) &= x^2 - x, \\ E_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{1}{4}, \\ E_4(x) &= x^4 - 2x^3 + x. \end{aligned}$$

These Euler polynomials should not be confused with the Eulerian polynomials of the preceding part.

Theorem 20.2 (Basic Euler-polynomial laws). *For $n \geq 0$,*

$$E'_n(x) = nE_{n-1}(x), \quad (20.2)$$

$$E_n(x+y) = \sum_{k=0}^n \binom{n}{k} E_k(x) y^{n-k}, \quad (20.3)$$

$$E_n(x+1) + E_n(x) = 2x^n, \quad (20.4)$$

$$E_n(1-x) = (-1)^n E_n(x). \quad (20.5)$$

Consequently,

$$2 \sum_{r=0}^{N-1} (-1)^r (x+r)^n = E_n(x) + (-1)^{N-1} E_n(x+N). \quad (20.6)$$

Proof. The derivative and addition laws follow from (20.1) exactly as for any Appell sequence. Adding the generating functions at x and $x+1$ gives $2e^{xt}$, proving (20.4). Replacing t by $-t$ gives the reflection law. Finally, apply (20.4) at $x+r$, multiply by $(-1)^r$, and sum; all interior Euler-polynomial terms cancel. $\square$

Theorem 20.3 (Bernoulli–Euler connection). *For $n \geq 0$,*

$$E_n(x) = \frac{2^{n+1}}{n+1} \left[\beta_{n+1}\left(\frac{x+1}{2}\right) - \beta_{n+1}\left(\frac{x}{2}\right) \right]. \quad (20.7)$$

Proof. Multiply the right side by $t^n/n!$ and sum over n . After putting $m = n + 1$, the result is

$$\begin{aligned} \frac{1}{t} \left[\frac{2te^{(x+1)t}}{e^{2t} - 1} - \frac{2te^{xt}}{e^{2t} - 1} \right] &= \frac{2e^{xt}(e^t - 1)}{e^{2t} - 1} \\ &= \frac{2e^{xt}}{e^t + 1}, \end{aligned}$$

which is the defining generating function. $\square$

The centered values produce the classical signed secant numbers. Define $\mathcal{E}_n = 2^n E_n(1/2)$. Then

$$\sum_{n \geq 0} \mathcal{E}_n \frac{z^n}{n!} = \operatorname{sech} z. \quad (20.8)$$

Hence $\mathcal{E}_{2m+1} = 0$ and $\mathcal{E}_0, \mathcal{E}_2, \mathcal{E}_4, \dots = 1, -1, 5, -61, 1385, \dots$. This follows by setting $x = 1/2$ in (20.1) and then replacing t by $2z$.

20.2 Genocchi polynomials and numbers

Definition 20.4. Define Genocchi polynomials by

$$\frac{2te^{xt}}{e^t + 1} = \sum_{n \geq 0} G_n(x) \frac{t^n}{n!}, \quad G_n := G_n(0). \quad (20.9)$$

Theorem 20.5. *For $n \geq 1$,*

$$G_n(x) = nE_{n-1}(x), \quad (20.10)$$

$$G_n = 2(1 - 2^n)\beta_n, \quad (20.11)$$

$$G_n(x+1) + G_n(x) = 2nx^{n-1}. \quad (20.12)$$

Thus $G_0 = 0, G_1 = 1, G_2 = -1, G_3 = 0, G_4 = 1, G_6 = -3, G_8 = 17$.

Proof. Multiplying the Euler-polynomial generating function by t proves (20.10). For (20.11), use

$$\frac{2t}{e^t + 1} = \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1} = 2 \sum_{n \geq 0} \beta_n \frac{t^n}{n!} - 2 \sum_{n \geq 0} \beta_n \frac{(2t)^n}{n!}.$$

Coefficient comparison gives the result. The difference identity follows by adding the generating functions at x and $x + 1$. $\square$

Chapter 21

Bernoulli polynomials of the second kind

These polynomials are also called Cauchy polynomials of the first kind. Their natural delta series is $\log(1+t)$ rather than t , so they belong to the Sheffer rather than the Appell class.

Definition 21.1. Define $b_n(x)$ by

$$\frac{t(1+t)^x}{\log(1+t)} = \sum_{n \geq 0} b_n(x) \frac{t^n}{n!}, \quad b_n := b_n(0). \quad (21.1)$$

The first polynomials are

$$\begin{aligned} b_0(x) &= 1, \\ b_1(x) &= x + \frac{1}{2}, \\ b_2(x) &= x^2 - \frac{1}{6}, \\ b_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{1}{4}, \\ b_4(x) &= x^4 - 4x^3 + 4x^2 - \frac{19}{30}. \end{aligned}$$

Theorem 21.2 (Integral and finite-difference representations). *For $n \geq 0$,*

$$b_n(x) = \int_0^1 x + u^n \, du, \quad (21.2)$$

$$b_n(x+1) - b_n(x) = nx^{n-1}, \quad (21.3)$$

$$b'_n(x) = nx^{n-1}, \quad (21.4)$$

$$b_n(x) = b_n + n \sum_{k=1}^n \frac{s(n-1, k-1)}{k} x^k, \quad (21.5)$$

$$b_n\left(\frac{n}{2} - 1 - x\right) = (-1)^n b_n\left(\frac{n}{2} - 1 + x\right). \quad (21.6)$$

Proof. Since $\sum_{n \geq 0} x + u^n t^n / n! = (1+t)^{x+u}$, integration over $0 \leq u \leq 1$ gives

$$\int_0^1 (1+t)^{x+u} \, du = (1+t)^x \frac{t}{\log(1+t)},$$

proving (21.2). Subtracting the generating functions at $x + 1$ and x gives $t(1 + t)^x$, which proves the difference law. Differentiating with respect to x cancels the logarithm and gives the same right side, proving (21.4). Since $x^{n-1} = \sum_{j=0}^{n-1} s(n-1, j)x^j$, integration from 0 to x gives (21.5). Finally use

$$z^n = (-1)^n n - 1 - z^n$$

in the integral representation and change u to $1 - u$. □

The inverse relation between $\log(1 + t)$ and $e^t - 1$ gives the diagonal coefficient

$$[t^{n-1}] \left(\frac{t}{\log(1 + t)} \right)^n = \frac{1}{(n-1)!}, \quad (21.7)$$

which will reappear as a direct Lagrange-inversion example. The polynomials $b_n(x)$ will also provide a concrete Sheffer sequence in the final synthesis part.

Chapter 22

Euler–Maclaurin summation and Darboux–Taylor remainders

22.1 Periodic Bernoulli functions

For $r \geq 1$, put $\tilde{\beta}_r(x) = \beta_r(x - \lfloor x \rfloor)$ away from integers. At integers the value of $\tilde{\beta}_1$ may be chosen arbitrarily in integration formulas; the symmetric convention sets it to 0. For $r \geq 2$ the function is continuous and 1-periodic. On every open unit interval,

$$\frac{d}{dx} \tilde{\beta}_r(x) = r \tilde{\beta}_{r-1}(x). \quad (22.1)$$

Theorem 22.1 (Euler–Maclaurin with an exact remainder). *Let $a < b$ be integers, let $m \geq 1$, and let $f \in C^{2m}([a, b])$. Then*

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(x) \, dx + \frac{f(a) + f(b)}{2} \\ &\quad + \sum_{r=1}^m \frac{\beta_{2r}}{(2r)!} \left(f^{(2r-1)}(b) - f^{(2r-1)}(a) \right) + R_m, \end{aligned} \quad (22.2)$$

$$R_m = - \frac{1}{(2m)!} \int_a^b \tilde{\beta}_{2m}(x) f^{(2m)}(x) \, dx. \quad (22.3)$$

Moreover,

$$|R_m| \leq \frac{2\zeta(2m)}{(2\pi)^{2m}} \int_a^b |f^{(2m)}(x)| \, dx. \quad (22.4)$$

Proof. On one interval $[k, k+1]$, integration by parts gives

$$\int_k^{k+1} \tilde{\beta}_1(x) f'(x) \, dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x) \, dx.$$

Summing from $k = a$ to $b-1$ and then adding $(f(a) + f(b))/2$ gives

$$\sum_{k=a}^b f(k) = \int_a^b f(x) \, dx + \frac{f(a) + f(b)}{2} + \int_a^b \tilde{\beta}_1(x) f'(x) \, dx. \quad (22.5)$$

Use (22.1) and integrate by parts successively. Odd Bernoulli numbers beyond β_1 vanish, so the boundary terms occur only at even indices and are exactly the displayed sum; stopping after $2m$ derivatives leaves (22.3). Finally, Theorem 19.8 gives

$$\sup_x |\tilde{\beta}_{2m}(x)| \leq \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m),$$

and insertion in the remainder integral proves (22.4). $\square$

22.2 Power sums, harmonic numbers, and factorials

For a polynomial f , the Euler–Maclaurin process terminates and is equivalent to Faulhaber’s formula. For nonpolynomial f it produces asymptotic expansions with computable remainders.

Corollary 22.2. *For every fixed $m \geq 1$,*

$$H_N = \log N + \gamma + \frac{1}{2N} - \sum_{r=1}^{m-1} \frac{\beta_{2r}}{2r N^{2r}} + O(N^{-2m}) \quad (N \rightarrow \infty). \quad (22.6)$$

Proof. Apply Theorem 22.1 to $f(x) = 1/x$ on $[N, M]$, then let $M \rightarrow \infty$. Every differentiated term at M vanishes, and the remainder bound is $O(N^{-2m})$. The constant left after subtracting $\log N$ is, by definition, Euler’s constant γ . $\square$

Corollary 22.3 (Stirling expansion). *For every fixed $m \geq 1$,*

$$\begin{aligned} \log N! = \left(N + \frac{1}{2}\right) \log N - N + \frac{1}{2} \log(2\pi) \\ + \sum_{r=1}^{m-1} \frac{\beta_{2r}}{2r(2r-1)N^{2r-1}} + O(N^{-(2m-1)}). \end{aligned} \quad (22.7)$$

Proof. Apply Euler–Maclaurin to $f(x) = \log x$ and absorb the lower-end terms into a constant. The derivatives $f^{(j)}(x) = (-1)^{j-1}(j-1)!x^{-j}$ give the displayed inverse powers and the remainder estimate. The constant is $\frac{1}{2} \log(2\pi)$, as follows from Wallis’ product (equivalently, from the central-binomial-coefficient estimate obtained from that product). This is the usual proof of Stirling’s formula with all higher corrections retained. $\square$

22.3 Darboux’s repeated-integration formula

The Darboux item in the supplementary material is the integration-by-parts mechanism behind Taylor expansion. In a normalization convenient here it is the following exact identity.

Theorem 22.4 (Taylor formula with integral remainder). *If $f \in C^m$ on the segment joining a and x , then*

$$f(x) = \sum_{r=0}^{m-1} \frac{f^{(r)}(a)}{r!} (x-a)^r + \frac{1}{(m-1)!} \int_a^x (x-t)^{m-1} f^{(m)}(t) dt. \quad (22.8)$$

Proof. For $m = 1$ this is the fundamental theorem of calculus. Integrate the remainder by parts, taking $u = f^{(m)}(t)$ and $dv = (x-t)^{m-1} dt / (m-1)!$. The boundary at $t = x$ vanishes and the boundary at $t = a$ contributes $f^{(m-1)}(a)(x-a)^{m-1} / (m-1)!$; the remaining integral is the remainder of order $m-1$. Induction gives the formula. $\square$

Both Theorems 22.1 and 22.4 are repeated-integration identities. Taylor’s kernel is an ordinary power, whereas Euler–Maclaurin’s kernel is a periodic Bernoulli polynomial adapted to summation over a lattice.

Chapter 23

Pochhammer symbols, polygamma functions, and Barnes' G -function

The supplementary MathWorld material connects the factorial bases of the first parts with gamma quotients and their parameter derivatives. Bell polynomials then reappear as the universal coefficients for differentiating those quotients. Gamma, polygamma, and Barnes normalizations are cross-checked against the DLMF [13].

23.1 Pochhammer symbols and the binomial series

For $n \in \mathbb{N}$, the rising Pochhammer symbol is

$$(z)_n = z^{\overline{n}} = z(z+1) \cdots (z+n-1), \quad (z)_0 = 1.$$

Where the quotient is defined,

$$(z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \quad (23.1)$$

This follows immediately by iterating $\Gamma(z+1) = z\Gamma(z)$ and then holds meromorphically in z . More generally one may define

$$(z)_\alpha = \frac{\Gamma(z+\alpha)}{\Gamma(z)} \quad (23.2)$$

for arbitrary complex α away from poles.

Theorem 23.1 (Generalized binomial theorem). *As a formal power series, and analytically for $|t| < 1$,*

$$(1-t)^{-z} = \sum_{n \geq 0} \frac{(z)_n}{n!} t^n. \quad (23.3)$$

Consequently,

$$(x+y)_n = \sum_{k=0}^n \binom{n}{k} (x)_k (y)_{n-k}. \quad (23.4)$$

Proof. Let $A(t) = \sum a_n t^n$ be the formal solution of $(1-t)A'(t) = zA(t)$ with $A(0) = 1$. Coefficient comparison gives $(n+1)a_{n+1} = (z+n)a_n$, hence $a_n = (z)_n/n!$. The analytic solution of the same initial-value problem is $(1-t)^{-z}$, proving (23.3). Multiply the series at parameters x and y and compare the coefficient of t^n to obtain (23.4). $\square$

Since $(z)_n = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} z^k$, the unsigned first-kind Stirling triangle is exactly the monomial expansion of the Pochhammer basis. The binomial identity above says that this basis is of binomial type.

23.2 Digamma and polygamma series

Define

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}, \quad \psi^{(m)}(z) = \frac{d^m}{dz^m} \psi(z).$$

The Weierstrass product

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right) e^{-z/k} \quad (23.5)$$

converges locally uniformly. Logarithmic differentiation therefore gives, for z away from the nonpositive integers,

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+z} \right). \quad (23.6)$$

For $m \geq 1$ and $\Re z > 0$, termwise differentiation yields

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{m+1}}. \quad (23.7)$$

The differentiated series converges locally uniformly in the right half-plane, which justifies every derivative. The recurrence

$$\psi^{(m)}(z+1) - \psi^{(m)}(z) = (-1)^m \frac{m!}{z^{m+1}} \quad (23.8)$$

follows either from the series or by differentiating $\psi(z+1) = \psi(z) + 1/z$.

At positive integers these formulas become

$$\psi(n+1) = H_n - \gamma, \quad (23.9)$$

$$\psi^{(m)}(n+1) = (-1)^{m+1} m! \left(\zeta(m+1) - H_n^{(m+1)} \right), \quad (23.10)$$

where $H_n^{(r)} = \sum_{j=1}^n j^{-r}$. Thus harmonic-number formulas for first-kind Stirling diagonals are finite-difference shadows of polygamma identities.

23.3 Parameter derivatives of factorials

For $r \geq 1$, put

$$H_n^{(r)}(z) = \sum_{j=0}^{n-1} \frac{1}{(z+j)^r}. \quad (23.11)$$

Then (23.8) gives

$$\psi^{(r-1)}(z+n) - \psi^{(r-1)}(z) = (-1)^{r-1} (r-1)! H_n^{(r)}(z). \quad (23.12)$$

Theorem 23.2 (Bell-polynomial parameter derivatives). *For $m \geq 0$,*

$$\frac{d^m}{dz^m} (z)_n = (z)_n \mathcal{B}_m(L_1, L_2, \dots, L_m), \quad (23.13)$$

where

$$L_r = (-1)^{r-1} (r-1)! H_n^{(r)}(z). \quad (23.14)$$

More generally, for the generalized symbol $(z)_\alpha$ one replaces L_r by $\psi^{(r-1)}(z+\alpha) - \psi^{(r-1)}(z)$.

Proof. Write

$$(z)_n = \exp(\log \Gamma(z+n) - \log \Gamma(z)).$$

The r th derivative of the exponent is the polygamma difference in (23.12). The exponential specialization (24.5) now gives (23.13). The same argument with n replaced by α proves the generalized statement. $\square$

For example,

$$\frac{d}{dz}(z)_n = (z)_n H_n^{(1)}(z), \quad (23.15)$$

$$\frac{d^2}{dz^2}(z)_n = (z)_n \left(H_n^{(1)}(z)^2 - H_n^{(2)}(z) \right), \quad (23.16)$$

$$\frac{d^3}{dz^3}(z)_n = (z)_n \left(H_n^{(1)}(z)^3 - 3H_n^{(1)}(z)H_n^{(2)}(z) + 2H_n^{(3)}(z) \right). \quad (23.17)$$

The same calculation gives the finite parameter-shift expansion

$$\frac{(z+h)_n}{(z)_n} = \exp \left(\sum_{r=1}^n \frac{(-1)^{r-1}}{r} H_n^{(r)}(z) h^r \right), \quad (23.18)$$

valid formally in h and analytically until a shifted pole is encountered. Extracting powers of h again produces complete Bell polynomials.

23.4 Barnes' G -function

The Barnes function is the canonical interpolation of a product of factorials. Define it initially by the locally uniformly convergent Weierstrass product

$$G(z+1) = (2\pi)^{z/2} \exp \left(-\frac{1}{2}z(z+1) - \frac{1}{2}\gamma z^2 \right) \times \prod_{k=1}^{\infty} \left(1 + \frac{z}{k} \right)^k \exp \left(-z + \frac{z^2}{2k} \right). \quad (23.19)$$

Indeed,

$$k \log \left(1 + \frac{z}{k} \right) - z + \frac{z^2}{2k} = O(k^{-2})$$

locally uniformly in z , so the logarithm converges normally away from the zeros and the product defines an entire function.

Theorem 23.3 (Functional equation and factorial interpolation). *The product (23.19) satisfies*

$$G(z+1) = \Gamma(z)G(z), \quad G(1) = 1. \quad (23.20)$$

Consequently,

$$G(N+1) = \prod_{k=1}^{N-1} k! \quad (N \geq 1). \quad (23.21)$$

Proof. The value $G(1) = 1$ follows by setting $z = 0$ in the product. Divide the product at z by the one at $z-1$ and truncate at $k = N$. The algebraic part telescopes:

$$\prod_{k=1}^N \left(\frac{k+z}{k+z-1} \right)^k = \frac{(N+z)^N \Gamma(z)}{\Gamma(N+z)}. \quad (23.22)$$

Including the exponential factors gives

$$\frac{G(z+1)}{G(z)} = \Gamma(z) \sqrt{2\pi} e^{-z-\gamma z+\gamma/2} \times \lim_{N \rightarrow \infty} \frac{(N+z)^N}{\Gamma(N+z)} e^{-N+(z-1/2)H_N}.$$

Stirling's formula and $H_N = \log N + \gamma + o(1)$ show that the limit on the second line is

$$\frac{e^{z+\gamma(z-1/2)}}{\sqrt{2\pi}},$$

which cancels the preceding prefactor. Thus the ratio is $\Gamma(z)$. Iteration at positive integers proves (23.21). $\square$

Theorem 23.4 (Alexeiewsky–Barnes integral). *On any simply connected domain where consistent logarithm branches are chosen,*

$$\log G(z+1) = \frac{z}{2} \log(2\pi) - \frac{1}{2} z(z+1) + z \log \Gamma(z+1) - \int_0^z \log \Gamma(t+1) dt. \quad (23.23)$$

Proof. Logarithmic differentiation of (23.19), justified by normal convergence, gives

$$\frac{d}{dz} \log G(z+1) = \frac{1}{2} \log(2\pi) - z - \frac{1}{2} - \gamma z + \sum_{k \geq 1} \frac{z^2}{k(k+z)}.$$

From (23.6), $\psi(z+1) = -\gamma + \sum_{k \geq 1} z/[k(k+z)]$; hence the last two terms combine to $z\psi(z+1)$. The derivative of the right side of (23.23) is also $\frac{1}{2} \log(2\pi) - z - \frac{1}{2} + z\psi(z+1)$. Both sides vanish at $z = 0$, so they are equal. $\square$

23.5 Euler–Maclaurin asymptotics for Barnes G

Let A be the Glaisher–Kinkelin constant, characterized by

$$\log A = \lim_{N \rightarrow \infty} \left\{ \sum_{k=1}^N k \log k - \left(\frac{N^2}{2} + \frac{N}{2} + \frac{1}{12} \right) \log N + \frac{N^2}{4} \right\}. \quad (23.24)$$

Theorem 23.5 (Barnes asymptotic expansion). *For every fixed $m \geq 1$ and positive integers $N \rightarrow \infty$,*

$$\begin{aligned} \log G(N+1) &= \frac{N^2}{4} + N \log \Gamma(N+1) - \left(\frac{N(N+1)}{2} + \frac{1}{12} \right) \log N - \log A \\ &\quad + \sum_{r=1}^{m-1} \frac{\beta_{2r+2}}{2r(2r+1)(2r+2)N^{2r}} + O(N^{-2m}). \end{aligned} \quad (23.25)$$

Equivalently,

$$\begin{aligned} \log G(N+1) &= \frac{N^2}{2} \log N - \frac{3N^2}{4} + \frac{N}{2} \log(2\pi) - \frac{1}{12} \log N + \left(\frac{1}{12} - \log A \right) \\ &\quad + \sum_{r=1}^{m-1} \frac{\beta_{2r+2}}{4r(r+1)N^{2r}} + O(N^{-2m}). \end{aligned} \quad (23.26)$$

The constant $\frac{1}{12} - \log A$ is also $\zeta'(-1)$.

Proof. Apply Theorem 22.1 to $f(x) = x \log x$. Since $f^{(q)}(x) = (-1)^q (q-2)! x^{1-q}$ for $q \geq 2$, collecting the upper-end terms gives

$$\sum_{k=1}^N k \log k = \left(\frac{N^2}{2} + \frac{N}{2} + \frac{1}{12} \right) \log N - \frac{N^2}{4} + \log A - \sum_{r=1}^{m-1} \frac{\beta_{2r+2}}{2r(2r+1)(2r+2)N^{2r}} + O(N^{-2m}). \quad (23.27)$$

The existence of the constant and the stated error follow from the exact Euler–Maclaurin remainder bound. From (23.21), rearranging the triangular sum of logarithms gives the exact identity

$$\log G(N+1) = N \log \Gamma(N+1) - \sum_{k=1}^N k \log k.$$

Substitute (23.27) to obtain (23.25). Substitution of Stirling's expansion (22.7) and collection of equal inverse powers gives (23.26); the coefficient identity is

$$\frac{\beta_{2r+2}}{(2r+2)(2r+1)} + \frac{\beta_{2r+2}}{2r(2r+1)(2r+2)} = \frac{\beta_{2r+2}}{4r(r+1)}.$$

The equality $\zeta'(-1) = 1/12 - \log A$ follows by differentiating the Euler–Maclaurin continuation of the zeta function at $s = -1$. $\square$

The sectorial complex expansion has the same coefficients and holds away from the negative real axis. The integer proof above already exhibits the essential mechanism: Barnes G packages a second summation of $\log z$, so one more layer of Bernoulli corrections appears beyond Stirling's expansion for Γ .

Part VII

Differentiation, Composition, and Reversion

Chapter 24

Faà di Bruno's formula

24.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 24.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (24.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

24.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 24.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (24.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (24.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (24.1) gives (24.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (24.3). $\square$

For $n = 4$ this reads

$$(f \circ g)^{(4)} = f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2g'' + 3f''(g)(g'')^2 + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \quad (24.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

24.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (24.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (24.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

24.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial / \partial x_j$.

Theorem 24.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (24.6)$$

Proof. The proof of Theorem 24.1 applies verbatim, except that the new derivative is $\partial / \partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (24.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

24.5 Canonical outer functions

Faà di Bruno becomes a library of higher-derivative rules once the derivatives of the outer function are inserted.

Theorem 24.4 (Power, logarithm, reciprocal, and trigonometric rules). *For $n \geq 1$ and at points where*

the expressions are defined,

$$D^n g(x)^\alpha = \sum_{k=1}^n \alpha^k g(x)^{\alpha-k} \mathcal{B}_{n,k}(g', g'', \dots), \quad (24.7)$$

$$D^n \log g(x) = \sum_{k=1}^n (-1)^{k-1} (k-1)! \frac{\mathcal{B}_{n,k}(g', g'', \dots)}{g(x)^k}, \quad (24.8)$$

$$D^n \frac{1}{g(x)} = \sum_{k=1}^n (-1)^k k! \frac{\mathcal{B}_{n,k}(g', g'', \dots)}{g(x)^{k+1}}, \quad (24.9)$$

$$D^n \sin(g(x)) = \sum_{k=1}^n \sin\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}(g', g'', \dots), \quad (24.10)$$

$$D^n \cos(g(x)) = \sum_{k=1}^n \cos\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}(g', g'', \dots). \quad (24.11)$$

For $n = 0$, the original function is recovered separately.

Proof. In (24.3), insert respectively

$$D^k u^\alpha = \alpha^k u^{\alpha-k}, \quad D^k \log u = (-1)^{k-1} (k-1)! u^{-k},$$

$$D^k u^{-1} = (-1)^k k! u^{-k-1}, \quad D^k \sin u = \sin(u + k\pi/2), \quad D^k \cos u = \cos(u + k\pi/2).$$

No further combinatorial calculation is required. $\square$

24.6 Leibniz rules and higher quotients

Theorem 24.5 (Multinomial Leibniz rule). *For q sufficiently differentiable functions,*

$$D^n \prod_{r=1}^q f_r(x) = \sum_{j_1 + \dots + j_q = n} \binom{n}{j_1, \dots, j_q} \prod_{r=1}^q f_r^{(j_r)}(x). \quad (24.12)$$

Proof. Taylor-expand every factor at x :

$$\prod_{r=1}^q f_r(x+h) = \prod_{r=1}^q \sum_{j_r \geq 0} f_r^{(j_r)}(x) \frac{h^{j_r}}{j_r!}.$$

Extracting $n![h^n]$ gives (24.12). $\square$

Corollary 24.6 (Higher quotient rule). *If $g(x) \neq 0$, then*

$$D^n \frac{f(x)}{g(x)} = \sum_{r=0}^n \binom{n}{r} f^{(n-r)}(x) \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}(g', g'', \dots)}{g(x)^{k+1}}, \quad (24.13)$$

where $\mathcal{B}_{0,0} = 1$.

Proof. Apply the two-factor case of Theorem 24.5 to $f \cdot g^{-1}$ and use (24.9), including its $r = 0$ term. $\square$

24.7 Coordinate-free Faà di Bruno formula

Let E, F, G be Banach spaces, let $g : E \rightarrow F$ and $f : F \rightarrow G$, and write D^r for the symmetric r -linear Fréchet derivative.

Theorem 24.7 (Fréchet Faà di Bruno). *For directions $h_1, \dots, h_n \in E$,*

$$D^n(f \circ g)(x)[h_1, \dots, h_n] = \sum_{\pi \in \text{Part}([n])} D^{|\pi|}f(g(x)) \left[D^{|B|}g(x)[h_i : i \in B] \right]_{B \in \pi}. \quad (24.14)$$

One vector of F is supplied to $D^{|\pi|}f$ for each block B of π .

Proof. Induct on n . The case $n = 1$ is the Fréchet chain rule. Differentiate a term indexed by a partition π of $[n]$ in the new direction h_{n+1} . If the derivative strikes the outer multilinear map, the chain rule creates a new argument $Dg(x)[h_{n+1}]$, corresponding to the singleton block $\{n+1\}$. If it strikes the inner factor belonging to a block B , that factor becomes

$$D^{|B|+1}g(x)[h_i : i \in B, h_{n+1}],$$

which adjoins $n+1$ to B . Every partition of $[n+1]$ is obtained uniquely by deleting $n+1$ either from its singleton block or from its unique non-singleton block. Hence every term occurs exactly once. $\square$

When $E = F = G = \mathbb{R}$ and all directions equal 1, grouping partitions by their block-size multiplicities recovers (24.2). Coordinate indices are therefore bookkeeping added after the partition structure, not a new combinatorial theorem.

24.8 Derivatives of inverse functions

Let $y = f(x)$ with $f'(x) \neq 0$, and put $x = g(y) = f^{-1}(y)$.

Theorem 24.8 (Recursive inverse-derivative formula). *For $n \geq 2$,*

$$g^{(n)}(y) = -\frac{1}{f'(g(y))} \sum_{k=2}^n f^{(k)}(g(y)) \mathcal{B}_{n,k}(g'(y), g''(y), \dots, g^{(n-k+1)}(y)), \quad (24.15)$$

with $g'(y) = 1/f'(g(y))$. In particular, writing derivatives of f at $x = g(y)$,

$$g' = \frac{1}{f'}, \quad (24.16)$$

$$g'' = -\frac{f''}{(f')^3}, \quad (24.17)$$

$$g''' = \frac{3(f'')^2 - f'f'''}{(f')^5}, \quad (24.18)$$

$$g^{(4)} = \frac{-15(f'')^3 + 10f'f''f''' - (f')^2f^{(4)}}{(f')^7}. \quad (24.19)$$

Proof. For $n \geq 2$, differentiate $f(g(y)) = y$ n times and apply (24.3). The term with $k = 1$ is

$$f'(g(y))g^{(n)}(y).$$

Moving all terms with $k \geq 2$ to the other side gives (24.15). Substitution of the already computed lower derivatives gives the four displayed cases. The recursion is triangular because $\mathcal{B}_{n,k}$ with $k \geq 2$ involves derivatives of g of order at most $n-1$. $\square$

Lagrange inversion will replace this triangular recursion by a direct coefficient formula. The two descriptions are complementary: Faà di Bruno is local in derivative order, whereas Lagrange–Bürmann is global in the defining series.

24.9 Nested composition

For three scalar functions, repeated Faà di Bruno gives the explicit finite formula

$$D^n f(g(h(x))) = \sum_{k=0}^n f^{(k)}(g(h(x))) \mathcal{B}_{n,k}((g \circ h)', (g \circ h)'', \dots), \quad (24.20)$$

$$(g \circ h)^{(j)} = \sum_{\ell=0}^j g^{(\ell)}(h(x)) \mathcal{B}_{j,\ell}(h', h'', \dots). \quad (24.21)$$

Substituting the second line into the first is a closed formula involving only derivatives of f, g, h . Combinatorially, the inner partition groups derivative labels that strike one occurrence of h , and the outer partition groups those blocks according to occurrences of g . The canonical flattening of nested partitions is the combinatorial reason formal composition is associative.

Composition normalizations

For ordinary series, $\widehat{\mathcal{B}}_{n,k}$ extracts $[z^n]G(z)^k$; for exponential series, $\mathcal{B}_{n,k}$ extracts $n![z^n]G(z)^k/k!$; in derivative formulas the arguments are the actual derivatives $g^{(j)}(x)$. The conversion

$$\widehat{\mathcal{B}}_{n,k}(g_1, g_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!g_1, 2!g_2, \dots)$$

accounts for the factors $n!$, $k!$, and $j!$ that differ among the five drafts.

Chapter 25

Composition of ordinary and exponential series

25.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 25.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \dots b_{i_k}, \quad (25.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \dots b_n^{\pi_n} \quad (25.2)$$

$$= \sum_{k=1}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad (25.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \widehat{\mathcal{B}}_{n,k}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (25.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

25.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 25.2 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}). \quad (25.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (25.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp \left(\sum_{j \geq 1} a_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (25.7)$$

If the series converge, (25.5) is exactly (24.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 26

Lagrange inversion and Lagrange–Bürmann

26.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (26.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 26.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n]H(g(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n, \quad (26.2)$$

$$= [w^n]H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (26.3)$$

In particular,

$$[z^n]g(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n. \quad (26.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.2 give

$$\begin{aligned} [z^n]H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1}f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (26.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (26.3). Take $H(w) = w$ for (26.4). □

26.2 Analytic inverse functions

Theorem 26.2 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (26.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (26.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta = z - b$, and

$$\phi(u) = \frac{u}{f(a + u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta \phi(u)$. Formula (26.4), multiplied by $n!$, is exactly (26.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + O((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

26.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$\hat{f}_j = \frac{f_{j+1}}{(j+1)f_1}. \quad (26.7)$$

Theorem 26.3 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k n^{\bar{k}} \mathcal{B}_{n-1,k}(\hat{f}_1, \dots, \hat{f}_{n-k}). \quad (26.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} \hat{f}_j \frac{w^j}{j!}.$$

By (26.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1 + U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

26.4 Associahedra, plane trees, and signed face sums

The supplied drafts use both exponential and ordinary coefficient conventions near the associahedral formula. The geometric statement is an ordinary-series identity, so we isolate that normalization. Let

$$f(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad g(y) = f^{(-1)}(y) = y + \sum_{n \geq 1} b_n y^{n+1}. \quad (26.9)$$

Let K_n denote the associahedron of dimension $n - 1$. Its faces may be represented by plane rooted trees with $n + 1$ leaves and no unary internal vertices. If such a tree has m_j internal vertices of arity $j + 1$, put

$$k = \sum_{j \geq 1} m_j, \quad \sum_{j \geq 1} j m_j = n, \quad a_F = \prod_{j \geq 1} a_j^{m_j}. \quad (26.10)$$

The corresponding face has dimension $n - k$.

Lemma 26.4 (Plane-tree profile count). *For a profile $(m_1, m_2, \dots)$ satisfying (26.10), the number of plane rooted trees with that profile and $n + 1$ leaves is*

$$\frac{(n + k)!}{(n + 1)! \prod_{j \geq 1} m_j!}. \quad (26.11)$$

Proof. List the vertices in preorder and record $d(v) - 1$, where $d(v)$ is the number of children. A leaf contributes -1 , while an internal vertex of arity $j + 1$ contributes j . The word has $n + k + 1$ entries, total sum -1 , and multiplicities $n + 1, m_1, m_2, \dots$. The cycle lemma [21] says that exactly one cyclic shift of each such word has all proper partial sums nonnegative; that shift is precisely the preorder degree word of a plane rooted tree. Because the total sum is -1 , no word has a nontrivial period, so every cyclic orbit has size $n + k + 1$. Hence the number of trees is

$$\frac{1}{n + k + 1} \binom{n + k + 1}{n + 1, m_1, m_2, \dots} = \frac{(n + k)!}{(n + 1)! \prod_j m_j!}.$$

□

Theorem 26.5 (Associahedral Lagrange inversion). *For every $n \geq 1$,*

$$b_n = \sum_{F \text{ a face of } K_n} (-1)^{n - \dim F} a_F, \quad (26.12)$$

$$= \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum j m_j = n}} (-1)^k \frac{(n + k)!}{(n + 1)! \prod_{j \geq 1} m_j!} \prod_{j \geq 1} a_j^{m_j}, \quad k = \sum_j m_j. \quad (26.13)$$

Proof. Write $f(x) = xA(x)$, where $A(x) = 1 + \sum_{j \geq 1} a_j x^j$. The inverse equation is $g = y/A(g)$, so Lagrange inversion gives

$$b_n = [y^{n+1}]g(y) = \frac{1}{n + 1} [x^n]A(x)^{-(n+1)}.$$

The generalized binomial theorem and the multinomial theorem give

$$\begin{aligned} b_n &= \frac{1}{n + 1} \sum_{k=0}^n (-1)^k \binom{n + k}{k} [x^n] \left(\sum_{j \geq 1} a_j x^j \right)^k \\ &= \sum_{\substack{\sum j m_j = n \\ \sum m_j = k}} (-1)^k \frac{(n + k)!}{(n + 1)! \prod_j m_j!} \prod_j a_j^{m_j}, \end{aligned}$$

which is (26.13). By Theorem 26.4, the unsigned coefficient counts the trees of that profile. The standard face-tree correspondence identifies these trees with faces of K_n ; contraction of an internal edge merges two internal vertices, and a tree with k internal vertices represents a face of dimension $n - k$. Thus $(-1)^k = (-1)^{n-\dim F}$, proving (26.12). $\square$

Corollary 26.6 (The face numbers of the associahedron). *The number $f_d(K_n)$ of d -dimensional faces, $0 \leq d \leq n - 1$, is*

$$f_d(K_n) = \frac{1}{n+1} \binom{n-1}{d} \binom{2n-d}{n}. \quad (26.14)$$

In particular, K_n has the Catalan number $\frac{1}{n+1} \binom{2n}{n}$ of vertices, and $\sum_d (-1)^d f_d(K_n) = 1$.

Proof. A d -face has $k = n - d$ internal vertices. Summing (26.11) over profiles with $\sum m_j = k$ uses

$$\sum_{\substack{\sum j m_j = n \\ \sum m_j = k}} \frac{1}{\prod_j m_j!} = \frac{1}{k!} [x^n] \left(\frac{x}{1-x} \right)^k = \frac{1}{k!} \binom{n-1}{k-1}.$$

Substitution and $k = n - d$ give (26.14). The vertex case is $d = 0$. Finally set every $a_j = 1$ in (26.12). Then $f(x) = x/(1-x)$ and $g(y) = y/(1+y)$, so $b_n = (-1)^n$; multiplying the resulting identity by $(-1)^n$ gives the Euler relation. $\square$

The first inverse polynomials are

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_1a_2 - 5a_1^3, \\ b_4 &= -a_4 + 6a_1a_3 + 3a_2^2 - 21a_1^2a_2 + 14a_1^4. \end{aligned}$$

For K_4 , the coefficients 1, 6, 3, 21, 14 enumerate the whole polytope, the two product types of facets, the edges, and the vertices. The alternating face sum is therefore not a metaphor for inversion: it is the inverse coefficient polynomial itself.

Chapter 27

Algebraic and transcendental examples

27.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (27.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 27.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (27.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (27.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (26.4),

$$[z^n]x(z) = \frac{1}{n}[w^{n-1}](1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

27.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 27.2. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + O(z^5), \quad (27.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (26.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

27.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (27.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + O(\xi^6). \quad (27.6)$$

Proof. The derivative at $u = 0$ of the left side of (27.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2, c_2 = 1/16, c_3 = -1/192, c_4 = -1/3072, c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143\dots$

27.4 Binary trees and Catalan numbers

Let C_n count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (27.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n = [z^n]C(z) = \frac{1}{n}[w^{n-1}](1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (27.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n is used here to avoid confusing these binary-tree numbers with Bell numbers.

27.5 Inverse trigonometric coefficient identities

Inverse functions can evaluate coefficient extractions that are otherwise surprisingly opaque. Since

$$\frac{d}{dy} \arcsin y = (1 - y^2)^{-1/2} = \sum_{m \geq 0} \binom{2m}{m} \frac{y^{2m}}{4^m},$$

integration gives

$$\arcsin y = \sum_{m \geq 0} \frac{1}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}. \quad (27.9)$$

On the other hand, Lagrange inversion applied to $f(x) = \sin x$ gives

$$[y^{2m+1}] \arcsin y = \frac{1}{2m+1} [x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1}.$$

Comparison proves

$$\boxed{[x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1} = \frac{1}{4^m} \binom{2m}{m}}. \quad (27.10)$$

Likewise, $\arctan y = \sum_{m \geq 0} (-1)^m y^{2m+1} / (2m+1)$ and inversion of $f(x) = \tan x$ give

$$\boxed{[x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1} = (-1)^m.} \quad (27.11)$$

Both proofs are complete because the elementary derivative expansions identify the inverse coefficients, while Theorem 26.1 identifies the same coefficients with the displayed high powers.

27.6 Exponential and logarithmic diagonal identities

The inverse pair $f(x) = e^x - 1$ and $g(y) = \log(1 + y)$ gives

$$[x^{n-1}] \left(\frac{x}{e^x - 1} \right)^n = (-1)^{n-1}, \quad n \geq 1. \quad (27.12)$$

Indeed, (26.4) says that the left side divided by n is $[y^n] \log(1 + y) = (-1)^{n-1}/n$. In the notation of Equation (19.10), this is precisely $\beta_{n-1}^{(n)} = (-1)^{n-1}(n-1)!$.

Conversely, $f(x) = \log(1 + x)$ and $g(y) = e^y - 1$ give

$$\boxed{[x^{n-1}] \left(\frac{x}{\log(1 + x)} \right)^n = \frac{1}{(n-1)!}.} \quad (27.13)$$

Here the left side divided by n equals $[y^n](e^y - 1) = 1/n!$. This evaluates a diagonal of the higher powers of the Cauchy-number generating function from Equation (21.1).

27.7 Kepler's equation and the Fourier–Bessel resummation

A classical analytic application of reversion is Kepler's equation

$$E = M + e \sin E, \quad 0 \leq e < 1. \quad (27.14)$$

Taking $x = M$, $y = e$, and $f(u) = \sin u$ in (28.3) gives the convergent reversion series

$$E = M + \sum_{k \geq 1} \frac{e^k}{k!} \left(\frac{d}{dM} \right)^{k-1} \sin^k M. \quad (27.15)$$

It admits a more useful harmonic resummation.

Theorem 27.3 (Fourier–Bessel solution of Kepler's equation). *For $0 \leq e < 1$, the unique real solution of (27.14) is*

$$E = M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM), \quad (27.16)$$

where

$$J_n(t) = \frac{1}{2\pi} \int_0^{2\pi} \cos(nu - t \sin u) \, du. \quad (27.17)$$

The series converges absolutely and uniformly in M for every fixed $e < 1$, and uniformly for e in compact subsets of $[0, 1)$.

Proof. The map $E \mapsto M = E - e \sin E$ has derivative $1 - e \cos E \geq 1 - e > 0$, so it is strictly increasing and has a unique analytic inverse satisfying $E(M + 2\pi) = E(M) + 2\pi$. Thus $q(M) = E(M) - M$ is a real-analytic, odd, 2π -periodic function. Its Fourier series contains only sine terms and converges absolutely and uniformly.

Let c_n be its n th sine coefficient. Since $q = e \sin E$ and $dM = (1 - e \cos E) dE$,

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^{2\pi} q(M) \sin(nM) dM \\ &= \frac{e}{\pi} \int_0^{2\pi} \sin E \sin(n(E - e \sin E))(1 - e \cos E) dE. \end{aligned}$$

Now

$$\frac{d}{dE} \cos(n(E - e \sin E)) = -n(1 - e \cos E) \sin(n(E - e \sin E)).$$

Integration by parts, with vanishing boundary term, gives

$$c_n = \frac{e}{\pi n} \int_0^{2\pi} \cos E \cos(nE - ne \sin E) dE. \quad (27.18)$$

From (27.17) and $2 \cos E \cos \theta = \cos(\theta + E) + \cos(\theta - E)$, this integral is $\pi(J_{n-1}(ne) + J_{n+1}(ne))$. The recurrence

$$J_{n-1}(t) + J_{n+1}(t) = \frac{2n}{t} J_n(t) \quad (27.19)$$

follows directly from the integral: integrate $\frac{d}{du} \sin(nu - t \sin u)$ over a period to obtain

$$n \int_0^{2\pi} \cos(nu - t \sin u) du = t \int_0^{2\pi} \cos u \cos(nu - t \sin u) du.$$

Substitution in (27.18) yields $c_n = 2J_n(ne)/n$, proving (27.16). Uniform convergence on compact e -intervals also follows directly from analytic dependence of the periodic inverse, or from standard contour shifts in its Fourier coefficients. $\square$

Expanding the exponential in the integral gives the familiar entire series

$$J_n(t) = \sum_{r \geq 0} \frac{(-1)^r}{r!(n+r)!} \left(\frac{t}{2}\right)^{n+2r} \quad (n \geq 0). \quad (27.20)$$

One proof is to extract the constant Fourier coefficient of $e^{inu} \exp\{\frac{t}{2}(e^{-iu} - e^{iu})\}$; matching equal powers of e^{iu} and e^{-iu} gives exactly the displayed sum. Hence every term in (27.16) can itself be expanded into the perturbation series (27.15); the two formulas are different reorganizations of the same analytic inverse.

Chapter 28

Lagrange's reversion theorem in shifted form

28.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (28.1)$$

Theorem 28.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (28.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (28.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (26.2) to $H(u) = g(x + u)$:

$$[y^k]g(v) = \frac{1}{k} [u^{k-1}]g'(x + u)f(x + u)^k = \frac{1}{k!} D_x^{k-1}(g'(x)f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

28.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x)g(z)(1 - yf'(z)) \, dz. \quad (28.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (28.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (28.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (28.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 29

Multivariate Lagrange–Good inversion

One variable hides the Jacobian in an integration-by-parts identity. In several variables the Jacobian survives as a determinant and is the essential new feature. The determinant form developed here is Good’s multivariate extension [24]; its combinatorial meaning is illuminated by Gessel’s tree and cancellation methods [22].

29.1 Multivariate formal residues

For a Laurent series in d variables, define

$$\operatorname{Res}_{\mathbf{x}} H(\mathbf{x}) \, dx_1 \cdots dx_d = [x_1^{-1} \cdots x_d^{-1}] H(\mathbf{x}).$$

Lemma 29.1 (Formal multivariate change of variables). *Let $f_i(\mathbf{x}) = x_i u_i(\mathbf{x})$, where every $u_i(\mathbf{0})$ is invertible. Then*

$$\begin{aligned} \operatorname{Res}_{\mathbf{t}} H(\mathbf{t}) \, dt_1 \cdots dt_d \\ = \operatorname{Res}_{\mathbf{x}} H(\mathbf{f}(\mathbf{x})) \det \left(\frac{\partial f_i}{\partial x_j} \right)_{i,j=1}^d \, dx_1 \cdots dx_d. \end{aligned} \quad (29.1)$$

Proof. By linearity it is enough to consider the differential form $\mathbf{t}^{\mathbf{m}} dt_1 \wedge \cdots \wedge dt_d$. If some $m_j \neq -1$, that form is exact: up to the sign needed to move dt_j into position, it is

$$d \left(\frac{t_j^{m_j+1}}{m_j+1} \prod_{i \neq j} t_i^{m_i} dt_1 \wedge \cdots \widehat{dt_j} \cdots \wedge dt_d \right).$$

The residue of an exact formal top form is zero, because it is a coefficient of a formal partial derivative. Pullback commutes with exterior differentiation, so the residue after substitution is also zero.

It remains to consider $m_1 = \cdots = m_d = -1$. Since $f_i = x_i u_i$,

$$\frac{df_i}{f_i} = \frac{dx_i}{x_i} + \frac{du_i}{u_i}.$$

In the wedge product of these d forms, the product of all dx_i/x_i has residue 1. Every other term contains at least one regular form du_i/u_i , whose coefficients have no negative exponents; such a term cannot contain $x_1^{-1} \cdots x_d^{-1}$. Its residue is zero. Thus the substitution preserves the unique nonexact Laurent top form and annihilates all the others, proving the formula. $\square$

29.2 Good’s determinant formula

Let $\mathbf{w} = (w_1, \dots, w_d)$ be the unique vector of formal series satisfying

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0, \quad 1 \leq i \leq d. \quad (29.2)$$

Existence and uniqueness follow coefficient by coefficient: the constant term is zero, the coefficient of t_i is $\phi_i(\mathbf{0})$, and every coefficient of total degree N depends only on lower total degrees.

Theorem 29.2 (Lagrange–Good inversion). *For a multi-index $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{N}^d$ and any formal series F ,*

$$\begin{aligned} [t^{\mathbf{n}}]F(\mathbf{w}(\mathbf{t})) \\ = [\mathbf{x}^{\mathbf{n}}]F(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \end{aligned} \quad (29.3)$$

Proof. Represent the coefficient on the left as an iterated residue and make the substitution

$$t_i = \frac{x_i}{\phi_i(\mathbf{x})}.$$

Under this substitution, the solution $\mathbf{w}(\mathbf{t})$ becomes $\mathbf{x}$. The Jacobian entries are

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

The factors $t_i^{-n_i-1}$ contribute $x_i^{-n_i-1} \phi_i^{n_i+1}$, while factoring $1/\phi_i$ from each Jacobian row leaves $\phi_i^{n_i}$. We therefore obtain the coefficient on the right with determinant

$$\det \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

Let $D_x = \text{diag}(x_1, \dots, x_d)$, $D_\phi = \text{diag}(\phi_1, \dots, \phi_d)$, and $J_\phi = (\partial \phi_i / \partial x_j)$. The last matrix is $I - D_x D_\phi^{-1} J_\phi$. The identity $\det(I - AB) = \det(I - BA)$ changes its determinant to $\det(I - D_\phi^{-1} J_\phi D_x)$, whose (i, j) entry is the one in (29.3). The residue substitution is justified by Theorem 29.1. $\square$

Corollary 29.3 (Recovery of one-variable Lagrange–Bürmann). *For $d = 1$, (29.3) reduces to*

$$[t^n]F(w(t)) = [x^n]F(x)\phi(x)^n - [x^{n-1}]F(x)\phi(x)^{n-1}\phi'(x).$$

This equals

$$\frac{1}{n} [x^{n-1}]F'(x)\phi(x)^n. \quad (29.4)$$

Proof. Differentiate $F(x)\phi(x)^n$ and take the coefficient of x^{n-1} :

$$n[x^n]F\phi^n = [x^{n-1}]F'\phi^n + n[x^{n-1}]F\phi^{n-1}\phi'.$$

Rearrangement gives (29.4), which is (26.2). $\square$

29.3 A multitype tree specialization

Let $A = (a_{ij})$ be a $d \times d$ matrix of weights and consider

$$w_i = t_i \exp \left(\sum_{j=1}^d a_{ij} w_j \right). \quad (29.5)$$

This is the exponential generating system for rooted labelled trees with vertex types: a type- i root carries an unordered set of type- j subtrees, with edge weight a_{ij} . Good's formula gives, for every F ,

$$[t^n]F(\mathbf{w}) = [\mathbf{x}^n]F(\mathbf{x}) \exp \left(\sum_{i,j} n_i a_{ij} x_j \right) \times \det(\delta_{ij} - a_{ij} x_j)_{i,j=1}^d. \quad (29.6)$$

Indeed, $\partial_j \phi_i = a_{ij} \phi_i$, so the logarithmic Jacobian in (29.3) collapses to the displayed linear determinant. This example shows why a determinant, rather than an unwieldy alternating sum, is the natural multivariate correction.

29.4 Analytic convergence and critical sets

Formal inversion determines all coefficients without convergence assumptions. For analytic maps, the inverse centered at a regular point continues until it meets either a singularity of the defining map or the image of the critical set where the Jacobian determinant vanishes. In one variable this condition is $f'(w) = 0$; in the system (29.2) it is precisely the vanishing of the determinant in (29.3). Simple critical points generically create square-root Puiseux singularities and the familiar $n^{-3/2}$ coefficient factor. The branch point $-e^{-1}$ of Lambert W , the point $1/4$ of the Catalan generating function, and the points ± 1 of arcsin are the one-variable manifestations of this same geometric criterion.

Chapter 30

Laplace-type asymptotic integrals

30.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (30.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (30.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (30.3)$$

Let $\widehat{\mathcal{B}}_{k,j}$ denote the ordinary Bell polynomial.

Theorem 30.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (30.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (30.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = tA(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s))t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} [t^k] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (26.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k]A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (30.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 30.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VIII

Möbius, Sheffer, and Riordan Synthesis

Chapter 31

Möbius inversion on the partition lattice

Let Π_n be the lattice of partitions of $[n]$, ordered by refinement. The moment–cumulant formulas already exhibit its zeta transform; here we make the inverse explicit.

Theorem 31.1 (Möbius function of the partition lattice). *For $\pi \leq \sigma$ in Π_n , let m_C be the number of blocks of π contained in a block C of σ . Then*

$$\mu(\pi, \sigma) = \prod_{C \in \sigma} (-1)^{m_C-1} (m_C - 1)!. \quad (31.1)$$

In particular,

$$\mu(\pi, \hat{1}) = (-1)^{|\pi|-1} (|\pi| - 1)!. \quad (31.2)$$

Proof. The interval $[\pi, \sigma]$ is the direct product, over $C \in \sigma$, of the partition lattices on the m_C blocks of π lying in C . Möbius functions multiply on direct products, so it is enough to prove $\mu_{\Pi_m}(\hat{0}, \hat{1}) = (-1)^{m-1} (m - 1)!$ for every m .

We prove this by induction on m . If $\tau < \hat{1}$ is a partition of $[m]$, then $[\hat{0}, \tau]$ is the product of the smaller lattices $\Pi_{|B|}$ over $B \in \tau$. The induction hypothesis and multiplicativity therefore give

$$\mu_{\Pi_m}(\hat{0}, \tau) = \prod_{B \in \tau} (-1)^{|B|-1} (|B| - 1)!.$$

For a fixed τ , the absolute value on the right counts choices of a cyclic ordering on every block, hence permutations of $[m]$ whose cycle-support partition is τ ; its sign $(-1)^{m-|\tau|}$ is the sign of each such permutation. Thus the sum of the displayed candidates over all $\tau \in \Pi_m$ is the sum of the signs of all permutations in S_m . For $m > 1$ this sum is zero, since multiplication by $(1\ 2)$ is a sign-reversing involution. The defining recurrence $\sum_{\tau \leq \hat{1}} \mu(\hat{0}, \tau) = 0$ therefore forces the remaining top value to be the candidate $(-1)^{m-1} (m - 1)!$. The case $m = 1$ is immediate. Multiplicativity on $[\pi, \sigma]$ now proves (31.1). $\square$

Corollary 31.2 (Moments and cumulants as Möbius inverses). *If*

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad (31.3)$$

then

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|}. \quad (31.4)$$

Proof. Equation (31.3) is the zeta transform on Π_n of the multiplicative block weight κ . Apply Möbius inversion and use (31.2). Grouping partitions by their block-size profiles gives exactly (15.5) and (15.6). $\square$

The Boolean lattice controls inclusion–exclusion, while the partition lattice controls passage between arbitrary structures and connected components. Their Möbius functions explain why signs and factorials occur in the inverse binomial and inverse Bell transforms.

Chapter 32

Sheffer sequences and delta operators

The binomial-type families of Theorem 15.5 are the associated sequences of the wider Sheffer class. We use the exponential convention of Roman [15]; the group-theoretic identification with Riordan arrays is developed in [25].

Definition 32.1. Let $A(z)$ be a unit, $A(0) \neq 0$, and let $\beta(z) = \beta_1 z + \beta_2 z^2 + \cdots$ with $\beta_1 \neq 0$. The Sheffer sequence for (A, β) is defined by

$$\sum_{n \geq 0} s_n(x) \frac{z^n}{n!} = A(z) e^{x\beta(z)}. \quad (32.1)$$

The associated binomial-type sequence $p_n(x)$ has generating function $e^{x\beta(z)}$.

Theorem 32.2 (Sheffer identity and characterization). *The polynomials in (32.1) satisfy*

$$s_n(x+y) = \sum_{k=0}^n \binom{n}{k} s_k(x) p_{n-k}(y). \quad (32.2)$$

Conversely, if p_n is of binomial type and a polynomial sequence s_n satisfies (32.2), then its EGF has the form (32.1) for a unique unit $A(z)$.

Proof. Multiply the EGF $A(z)e^{x\beta(z)}$ by $e^{y\beta(z)}$ and compare coefficients to prove the identity. Conversely, let $S(x, z) = \sum s_n(x) z^n / n!$ and $P(y, z) = \sum p_n(y) z^n / n! = e^{y\beta(z)}$. The assumed identity is $S(x+y, z) = S(x, z)P(y, z)$. Setting $x = 0$ gives $S(y, z) = S(0, z)e^{y\beta(z)}$; hence $A(z) = S(0, z)$, uniquely. $\square$

Writing $A(z) = \sum a_j z^j / j!$ immediately gives

$$s_n(x) = \sum_{j=0}^n \binom{n}{j} a_j p_{n-j}(x). \quad (32.3)$$

Thus the unit A performs an invertible Appell convolution on an associated binomial-type family.

Let $\bar{\beta}$ be the compositional inverse of β . Functional calculus in the differentiation operator gives the delta operator

$$Q = \bar{\beta}(D_x). \quad (32.4)$$

Theorem 32.3 (Lowering law). *Both the associated and the Sheffer sequences satisfy*

$$Qp_n(x) = np_{n-1}(x), \quad Qs_n(x) = ns_{n-1}(x). \quad (32.5)$$

Proof. Since $D_x e^{x\beta(z)} = \beta(z) e^{x\beta(z)}$,

$$\bar{\beta}(D_x) e^{x\beta(z)} = \bar{\beta}(\beta(z)) e^{x\beta(z)} = z e^{x\beta(z)}.$$

The factor $A(z)$ is independent of x , so the same equation holds for the Sheffer EGF. Coefficient comparison gives both lowering laws. $\square$

Lagrange inversion is therefore internal to Sheffer calculus: it computes the coefficients of $\bar{\beta}$, and hence the differential operator that lowers the sequence.

32.1 Principal examples

The following table uses exactly one convention, so the comparisons are literal rather than up to rescaling.

Family	$(A(z), \beta(z))$	EGF and lowering operator
Monomials	$(1, z)$	$e^{xz}; Q = D.$
Falling factorials	$(1, \log(1+z))$	$(1+z)^x; Q = e^D - 1 = \Delta.$
Rising factorials	$(1, -\log(1-z))$	$(1-z)^{-x}; Q = 1 - e^{-D}.$
Touchard polynomials	$(1, e^z - 1)$	$e^{x(e^z-1)}; Q = \log(1+D).$
Bernoulli polynomials	$(z/(e^z - 1), z)$	$ze^{xz}/(e^z - 1); Q = D.$
Euler polynomials	$(2/(e^z + 1), z)$	$2e^{xz}/(e^z + 1); Q = D.$
Bernoulli polynomials of the second kind	$(z/\log(1+z), \log(1+z))$	$z(1+z)^x/\log(1+z); Q = e^D - 1.$

The first four rows are of binomial type. The last three have nontrivial prefactors and are genuine Sheffer sequences; the two Bernoulli families are adapted respectively to differentiation and finite differences.

Chapter 33

Exponential Riordan arrays

33.1 Definition, action, and group law

For a unit $g(z)$ and a delta series $f(z)$, define the lower-triangular array

$$R_{n,k} = \frac{n!}{k!} [z^n] g(z) f(z)^k. \quad (33.1)$$

If $a = (a_k)$ has EGF $A(z)$, then the transformed sequence $b_n = \sum_k R_{n,k} a_k$ has EGF

$$B(z) = g(z) A(f(z)). \quad (33.2)$$

Indeed, summing (33.1) against a_k gives $g(z) \sum_k a_k f(z)^k / k!$.

Theorem 33.1 (Riordan group law). *With arrays denoted by pairs,*

$$(g, f)(h, \ell) = (g h \circ f, \ell \circ f), \quad (33.3)$$

$$(g, f)^{-1} = \left(\frac{1}{g \circ \bar{f}}, \bar{f} \right), \quad (33.4)$$

where $\bar{f} = f^{\langle -1 \rangle}$.

Proof. Apply first (h, ℓ) and then (g, f) to an arbitrary EGF A :

$$A(z) \mapsto h(z) A(\ell(z)) \mapsto g(z) h(f(z)) A(\ell(f(z))),$$

which proves the product. Substitution of $(1/(g \circ \bar{f}), \bar{f})$ on either side gives $(1, z)$, the identity array. $\square$

The row polynomials of (A, β) are precisely the Sheffer sequence:

$$s_n(x) = \sum_{k=0}^n R_{n,k} x^k, \quad \sum_n s_n(x) \frac{z^n}{n!} = A(z) e^{x\beta(z)}. \quad (33.5)$$

Thus Sheffer composition is ordinary matrix multiplication in the exponential Riordan group.

33.2 Bell and Stirling arrays

The partial Bell-polynomial array is the Riordan array

$$\mathcal{B}_{n,k}(x_1, x_2, \dots) = \frac{n!}{k!} [z^n] \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right)^k. \quad (33.6)$$

The Stirling arrays are

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{k!} [z^n] (e^z - 1)^k, \quad (33.7)$$

$$s(n, k) = \frac{n!}{k!} [z^n] \log(1 + z)^k. \quad (33.8)$$

Since $e^z - 1$ and $\log(1 + z)$ are compositional inverses, their Riordan arrays are inverse matrices. This recovers Stirling inversion as a special case of (33.4); Faà di Bruno composition is the corresponding matrix product, and Lagrange inversion computes its inverse entries.

Theorem 33.2 (Connection constants for two Sheffer families). *Let s_n and r_n be Sheffer for (A, β) and (C, γ) , respectively, and let $\bar{\gamma}$ be the inverse of γ . In*

$$s_n(x) = \sum_{k=0}^n c_{n,k} r_k(x), \quad (33.9)$$

the connection coefficients are

$$c_{n,k} = \frac{n!}{k!} [z^n] \frac{A(z)}{C(\bar{\gamma}(\beta(z)))} (\bar{\gamma}(\beta(z)))^k. \quad (33.10)$$

Proof. In vector notation, $s = (A, \beta)x^\bullet$ and $r = (C, \gamma)x^\bullet$. Hence $s = (A, \beta)(C, \gamma)^{-1}r$. Use (33.3) and (33.4); the resulting Riordan pair is

$$\left(\frac{A(z)}{C(\bar{\gamma}(\beta(z)))}, \bar{\gamma}(\beta(z)) \right).$$

Its entry formula is exactly (33.10). □

This theorem systematically produces all change-of-basis triangles among monomials, factorials, Touchard, Bernoulli, Euler, and Cauchy polynomials. It also explains why apparently unrelated coefficient tables in the five drafts obey the same matrix laws.

33.3 Abel polynomials from tree reversion

For a parameter a , let $T_a(z)$ be the unique solution of

$$T_a(z) = ze^{aT_a(z)}. \quad (33.11)$$

Lagrange inversion gives

$$[z^n] e^{xT_a(z)} = \frac{x}{n} [u^{n-1}] e^{(x+an)u} = \frac{x(x+an)^{n-1}}{n!}. \quad (33.12)$$

Replacing a by $-a$ shows that the Abel polynomials

$$A_n(x) = x(x-an)^{n-1}, \quad A_0(x) = 1, \quad (33.13)$$

are of binomial type, with EGF $\exp(xT_{-a}(z))$. Therefore

$$\begin{aligned} (x+y)(x+y-an)^{n-1} \\ = \sum_{k=0}^n \binom{n}{k} x(x-ak)^{k-1} y(y-a(n-k))^{n-k-1}, \end{aligned} \quad (33.14)$$

with the endpoint factors interpreted by $A_0 = 1$. This is Abel's binomial identity, obtained without guessing: it is the binomial law forced by the tree reversion series.

Part IX

**Synthesis, Verification, and Source
Record**

Appendix A

Editorial synthesis of the five drafts

A.1 Method of consolidation

The five draft manuscripts are not independent treatments of five disjoint subjects. They are overlapping realizations of one coefficient calculus, with substantially different chapter orders, proof granularity, notation, and choices of examples. A literal concatenation would therefore repeat the elementary Stirling and Bell theory several times while separating formulas that are best understood as instances of one theorem. The consolidation used the following rules.

1. An identity appearing in more than one draft is stated once, in the strongest useful form, and is followed by one complete proof. Alternative arguments are retained when they expose genuinely different mechanisms, such as a bijection versus a formal-residue calculation.
2. Definitions are moved before every theorem that depends on them. Formal power-series statements are separated from analytic statements, so that convergence assumptions are neither silently imposed nor silently omitted.
3. Notational collisions are resolved globally. In particular, Bell numbers, Bernoulli numbers, Bell polynomials, Euler polynomials, Eulerian numbers, type- B Eulerian numbers, and binary-tree numbers are never denoted by the same undecorated symbol in the same context.
4. Schematic formulas are replaced by checkable formulas. Missing index ranges, boundary values, Jacobians, branch choices, and normalization factors are made explicit. Where an archived proof stopped after suggesting a pattern, the argument is completed by coefficient extraction, Möbius inversion, a cycle lemma, or formal residues.
5. Material in the supplementary Wikipedia and MathWorld folders is not copied as a second encyclopedia. It is absorbed into the existing dependency structure and proved using the same coefficient machinery.

A.2 Draft-by-draft concordance

Draft	Principal emphasis	Material retained in the unified text
<code>combinatorial_coefficient_calculus.tex</code>	The most developed dependency-driven treatment; strong coverage of Stirling, Bell, Eulerian, Faà di Bruno, Lagrange, congruence, and asymptotic methods.	Used as the structural spine. Its formal coefficient calculus, full triangular array theory, exact Bell-number analysis, Eulerian chapters, residue proof of Lagrange inversion, and Laplace coefficient formula are retained and expanded.
<code>Formulae_Compendium.tex</code>	A compact but wide survey, with particularly useful higher-derivative specializations and systematic reversion examples.	Supplies the expanded library of power, logarithm, reciprocal, trigonometric, product, quotient, Fréchet, and inverse-function derivative formulas, together with normalization cross-checks and the Kepler reversion example.
<code>Formulae_Monograph.tex</code> in <code>Formulae_Monograph_Package</code>	A proof-oriented monograph emphasizing transitions among factorial bases, labelled structures, special-function series, and certificate procedures.	Its alternative combinatorial proofs, generalized factorial triangles, finite-difference identities, special-function viewpoints, and exact low-order consistency checks are merged into the corresponding chapters.
<code>Formulae_Monograph.tex</code> in <code>Formulae_Monograph_Source_and_PDF</code>	A parallel monograph with a different order and additional attention to normalization, multivariate differentiation, and auditability.	Contributes the global notation policy, several repaired boundary conventions, the coordinate-free composition viewpoint, the formula-dependency map, and the source-audit methodology used in this appendix.
<code>Combinatorial_Formulae_and_Inversion_Theorems.tex</code>	A broad synthesis linking Stirling transforms, Bell polynomials, probability, inversion, implicit functions, and algebraic examples.	Contributes the Boolean- and partition-lattice perspectives, the expanded moment-cumulant discussion, inverse-series examples, and the bridge from one-variable inversion to Lagrange–Good inversion.

The entries above describe provenance, not mutually exclusive ownership. Many results occur in three or more drafts. Repetition of a theorem is removed, while distinct proofs, examples, or normalizations are retained in the theorem’s natural chapter.

A.3 Supplementary dossiers absorbed

Folder or item	Integrated contribution
<code>Wikipedia/Stirling_*</code>	Definitions, triangular recurrences, factorial-basis transitions, generating functions, finite differences, asymptotic regimes, and variants. These are unified in the Stirling parts and supplied with consistent signed/unsigned conventions.

Folder or item	Integrated contribution
Wikipedia/Bell_*	Set partitions, Touchard polynomials, Dobiński's formula, Bell-polynomial composition, congruences, and asymptotics. Complementary Bell numbers are added as the evaluation $T_n(-1)$.
Wikipedia/Eulerian_number.txt	Descent statistics, Worpitzky's identity, power sums, rational generating functions, and geometric interpretations, consolidated with the type- B and second-order variants.
Wikipedia/Fa*_di_Bruno's_formula.txt	Partition, multiplicity, Bell-polynomial, and multivariate forms of the higher chain rule, expanded here to Banach-space derivatives, higher quotients, and nested composition.
Wikipedia/Lagrange_*	Formal, analytic, shifted, and combinatorial inversion formulas, including the plane-tree and associahedral interpretation. The ordinary/exponential normalization ambiguity is explicitly resolved.
Wikipedia/Associahedron.*	Face-tree correspondence and face numbers. These are connected directly to the coefficient polynomial of a compositional inverse and proved using the cycle lemma and Lagrange inversion.
MathWorld Bernoulli, Euler, Genocchi, and Euler–Maclaurin items	Bernoulli and Euler Appell sequences, Nörlund polynomials, Cauchy–Bernoulli polynomials of the second kind, Genocchi numbers, modified Bernoulli coefficients, Fourier expansions, Faulhaber summation, and exact Euler–Maclaurin remainders.
MathWorld Pochhammer, digamma, polygamma, and Barnes items	Gamma-quotient factorials, parameter derivatives through complete Bell polynomials, harmonic and zeta evaluations, Barnes' functional equation, Alexeiewsky's integral, and the Euler–Maclaurin expansion of $\log G$.
MathWorld Darboux item	The repeated-integration form of Taylor's theorem and its relationship to the periodic-Bernoulli kernel in Euler–Maclaurin summation.

A.4 Externally researched extensions

The multivariate determinant formula is developed in the form introduced by Good and is connected with Gessel's combinatorial treatment [24, 22]. The final Sheffer–Riordan synthesis follows the formal group viewpoint of Roman and the explicit comparison of the Sheffer and Riordan groups by He, Hsu, and Shiue [15, 25]. Analytic normalizations for Bernoulli, gamma, polygamma, and Barnes functions were checked against the NIST Digital Library of Mathematical Functions [13]. These sources are used to extend and cross-check the supplied material; all identities needed in the body are proved there rather than delegated to a citation.

Appendix B

Normalization ledger

B.1 Ordinary versus exponential coefficients

Let

$$X(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}, \quad \hat{X}(z) = \sum_{j \geq 1} \hat{x}_j z^j.$$

The exponential and ordinary partial Bell polynomials are characterized by

$$\begin{aligned} \frac{X(z)^k}{k!} &= \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, x_2, \dots) \frac{z^n}{n!}, \\ \hat{X}(z)^k &= \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(\hat{x}_1, \hat{x}_2, \dots) z^n. \end{aligned}$$

Putting $x_j = j! \hat{x}_j$ gives the conversion

$$\hat{\mathcal{B}}_{n,k}(\hat{x}_1, \hat{x}_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1! \hat{x}_1, 2! \hat{x}_2, \dots). \quad (\text{B.1})$$

This single identity accounts for nearly every factorial discrepancy among the five drafts. It also explains why derivative formulas use exponential Bell polynomials, while ordinary Taylor substitution and the Laplace coefficients use ordinary Bell polynomials.

B.2 First-kind Stirling signs

The two first-kind conventions are

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}.$$

Consequently the inverse matrices are

$$(s(n, k))_{n, k \geq 0}^{-1} = \left(\begin{Bmatrix} n \\ k \end{Bmatrix} \right)_{n, k \geq 0},$$

whereas the unsigned array requires the sign diagonal:

$$\left(\begin{bmatrix} n \\ k \end{bmatrix} \right)_{n, k}^{-1} = (-1)^{n-k} \begin{Bmatrix} n \\ k \end{Bmatrix}.$$

Any formula moved between rising and falling factorials must carry this sign.

B.3 Bernoulli, Euler, and Eulerian objects

The Bernoulli convention is fixed by

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Euler polynomials are the Appell sequence generated by $2e^{xt}/(e^t + 1)$. Eulerian polynomials instead enumerate descents; they are not an Appell sequence. Type-*B* Eulerian numbers refer to the Coxeter group of signed permutations and are unrelated to Bernoulli notation. The disambiguation is semantic as well as typographical: these families diagonalize different operators.

B.4 Compositional inverses and branches

For a formal series $f(z) = a_1 z + O(z^2)$ with invertible a_1 , the notation $f^{\langle -1 \rangle}$ always means the unique formal compositional inverse with zero constant term. In analytic applications it means the germ selected by the stated base point. Global multivalued functions such as Lambert W , logarithms, and algebraic roots require a branch designation; a coefficient formula about one germ is not silently asserted on every branch.

B.5 Residues and Jacobians

In one variable,

$$\operatorname{Res}_t H(t) \, dt = \operatorname{Res}_x H(f(x)) f'(x) \, dx.$$

In several variables, f' becomes the full Jacobian determinant. The determinant in Good's theorem is therefore not an optional correction term: it is exactly the factor needed to make multivariate residue invariant under substitution.

Appendix C

Reproducible coefficient algorithms

C.1 Triangular arrays

For any target order N , the Stirling, Lah, and Eulerian triangles are generated by their proved two-term recurrences. A row-at-a-time implementation requires $O(N)$ memory and $O(N^2)$ ring operations. Exact integer arithmetic should be used whenever the inputs are integral; floating-point evaluation is unnecessary and obscures divisibility checks.

A generic lower-triangular recurrence has the form

$$a_{n,k} = u_{n,k}a_{n-1,k-1} + v_{n,k}a_{n-1,k}.$$

Initialize $a_{0,0} = 1$, declare out-of-range entries zero, and fill rows in increasing n . The boundary convention eliminates special branches in the inner loop and is part of the mathematical specification, not merely a programming convenience.

C.2 Partial Bell polynomials

The recurrence

$$\mathcal{B}_{n,k}(x_1, x_2, \dots) = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j \mathcal{B}_{n-j,k-1}(x_1, x_2, \dots)$$

computes the full table through order N . It follows by marking the block containing label 1, so every generated polynomial has a direct combinatorial certificate. Homogeneity provides two independent checks:

$$\begin{aligned} \mathcal{B}_{n,k}(cx_1, cx_2, \dots) &= c^k \mathcal{B}_{n,k}(x_1, x_2, \dots), \\ \mathcal{B}_{n,k}(cx_1, c^2x_2, \dots) &= c^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \end{aligned}$$

C.3 Formal composition and reversion

To compose truncated ordinary series modulo z^{N+1} , evaluate powers of the inner series only up to the degree needed by the current outer coefficient. Bell polynomials expose all coefficient dependencies explicitly. For reversion, Lagrange–Bürmann gives

$$[z^n]f^{(-1)}(z) = \frac{1}{n}[u^{n-1}] \left(\frac{u}{f(u)} \right)^n.$$

This is usually the clearest symbolic algorithm because every target coefficient is independent. Newton iteration

$$g_{\text{new}} = g - \frac{f(g) - z}{f'(g)}$$

doubles the number of correct coefficients at each step and is preferable at very high orders when fast multiplication is available. A reliable implementation computes by one method and verifies $f(g(z)) \equiv z \pmod{z^{N+1}}$ by the other.

C.4 Multivariate Good coefficients

For fixed multi-index $\mathbf{n}$, only monomials bounded componentwise by $\mathbf{n}$ are needed. Truncate every ϕ_i and F to that finite box, form the logarithmic Jacobian matrix

$$J_{ij} = \delta_{ij} - \frac{x_j}{\phi_i} \partial_j \phi_i,$$

compute its determinant in the truncated multivariate ring, multiply by $F \prod_i \phi_i^{n_i}$, and extract $\mathbf{x}^{\mathbf{n}}$. Substitution of the computed coefficients back into $w_i = t_i \phi_i(\mathbf{w})$ is an exact finite certificate.

C.5 Asymptotic coefficients

Euler–Maclaurin, saddle-point, and Laplace expansions should be stored as formal coefficient generators rather than as fixed decimal tables. For an expansion through order r , retain one additional derivative or coefficient and verify that it contributes only to the stated remainder order. The exact periodic-Bernoulli remainder in Euler–Maclaurin and the Cauchy bounds for analytic coefficient extraction provide quantitative, independently checkable error estimates.

Appendix D

Independent consistency checks

D.1 Low-order composition check

Writing

$$f(u) = f_0 + f_1 u + \frac{f_2}{2!} u^2 + \frac{f_3}{3!} u^3 + \frac{f_4}{4!} u^4 + \dots$$

and $g(x + h) = g_0 + g_1 h + g_2 h^2/2! + \dots$, direct multiplication through h^4 gives

$$D^4(f \circ g) = f_1 g_4 + f_2(4g_1 g_3 + 3g_2^2) + 6f_3 g_1^2 g_2 + f_4 g_1^4.$$

The coefficients 1, 4, 3, 6, 1 count set partitions of types 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1. Their sum is $B_4 = 15$, checking simultaneously the partition, multiplicity, and Bell-polynomial forms of Faà di Bruno.

D.2 Low-order inverse check

For

$$f(x) = x + a_1 x^2 + a_2 x^3 + a_3 x^4 + a_4 x^5 + O(x^6),$$

the associahedral formula gives

$$\begin{aligned} f^{(-1)}(y) = & y - a_1 y^2 + (2a_1^2 - a_2) y^3 + (5a_1 a_2 - 5a_1^3 - a_3) y^4 \\ & + (14a_1^4 - 21a_1^2 a_2 + 3a_2^2 + 6a_1 a_3 - a_4) y^5 + O(y^6). \end{aligned}$$

Direct substitution into $f(g(y))$ cancels each coefficient through y^5 . The absolute coefficients 1, 6, 3, 21, 14 in the fifth-order polynomial are the corresponding face-profile counts of K_4 .

D.3 Matrix check for Stirling and Riordan inversion

Truncating at any order N , multiply the signed first-kind matrix $(s(n, k))_{0 \leq n, k \leq N}$ by the second-kind matrix $(\{n_k\})_{0 \leq n, k \leq N}$. Factorial-basis substitution proves that the product is the identity over $\mathbb{Z}$; direct multiplication is an independent finite certificate. The same test applies to a general exponential Riordan pair: truncate (g, f) and $(1/(g \circ \bar{f}), \bar{f})$ and verify their product is $(1, z)$ modulo z^{N+1} .

D.4 Analytic versus formal checks

A formal identity can be checked in a finite quotient ring without evaluating a single transcendental function. An analytic identity needs a second layer: locate singularities and critical points, select branches, and bound the tail. Kepler's equation illustrates the distinction. The Lagrange series is formally forced by $E = M + e \sin E$; the inequality $1 - e \cos E > 0$ for $0 \leq e < 1$ proves the real analytic inverse, and the Fourier–Bessel series supplies a globally periodic resummation.

Appendix E

Formula-dependency map

The principal logical routes can be summarized without suppressing the parallel proof mechanisms:

Route	Dependency chain
Foundations	formal power series and residues $\longrightarrow$ factorial bases and finite differences $\longrightarrow$ Stirling and Lah transforms
Labelled structures	Boolean and partition Möbius inversion $\longrightarrow$ labelled sets and partitions $\longrightarrow$ Bell numbers and Bell polynomials
Descents and summation	Stirling transforms and finite differences $\longrightarrow$ Eulerian arrays and power sums $\longrightarrow$ Bernoulli–Euler summation calculus
Composition and inversion	Bell polynomials together with formal residues $\longrightarrow$ Faà di Bruno composition calculus $\longrightarrow$ Lagrange and Good inversion
Operator groups	factorial bases together with labelled-component calculus $\longrightarrow$ Sheffer sequences and Riordan groups $\longrightarrow$ substitution and inverse-array formulas
Special functions	Bernoulli–Euler calculus together with repeated composition $\longrightarrow$ gamma and polygamma identities $\longrightarrow$ Barnes’ G -function and asymptotic coefficient expansions

The map is not merely pedagogical. It predicts proof techniques. A problem about polynomial bases suggests triangular matrices or a delta operator; a problem about labelled components suggests exponential generating functions and Bell polynomials; an implicit substitution suggests residues and Lagrange inversion; a lattice sum suggests Bernoulli kernels and Euler–Maclaurin. Most formulas in the source corpus become short once moved to the coordinate system in which their operator is diagonal.

Appendix F

Completeness and correction record

F.1 Material intentionally deduplicated

Repeated introductions to ordinary and exponential generating functions, repeated first tables of Stirling and Bell numbers, and repeated statements of the same recurrence are represented once. Omitting duplicate prose does not omit a mathematical assertion: variants are recorded when their hypotheses, indexing, or proof mechanism differ.

F.2 Systematic corrections

The following classes of defect were repaired during consolidation.

1. Signed and unsigned first-kind Stirling numbers were separated whenever a rising-factorial formula had been combined with a falling-factorial sign.
2. The type- B descent index was fixed at positions $0, 1, \dots, n - 1$ with $\pi_0 = 0$, eliminating a one-step shift.
3. Ordinary inverse-series coefficients in the associahedral formula were separated from exponential Bell-polynomial normalizations.
4. Formal inversion hypotheses ($f(0) = 0$, invertible linear coefficient) were separated from analytic inverse-function hypotheses and branch choices.
5. The distributional reversion proof was supplied with its Jacobian and orientation condition.
6. Multivariate inversion was supplied with the full determinant that one variable hides.
7. Asymptotic displays with unnamed correction polynomials were replaced by coefficient recurrences or exact remainder formulas.
8. Special-function identities were assigned their actual pole exclusions, half-planes, sectors, or local convergence domains.

F.3 Coverage principle

Every distinct theorem, identity family, example class, normalization issue, and computational certificate found in the five drafts has been assigned a location in the unified dependency structure. The supplementary Wikipedia and MathWorld materials have been used to fill missing links and extend the theory, especially in Bernoulli–Euler summation, complementary Bell numbers, associahedra, factorial parameter derivatives, and Barnes’ G -function. The externally researched Good, Gessel, and Sheffer–Riordan material closes the progression from triangular one-variable transforms to multivariate and group-theoretic inversion.

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